Fundamentals of Geophysics, W. Lowrie

# 4.3 Geoelectricity

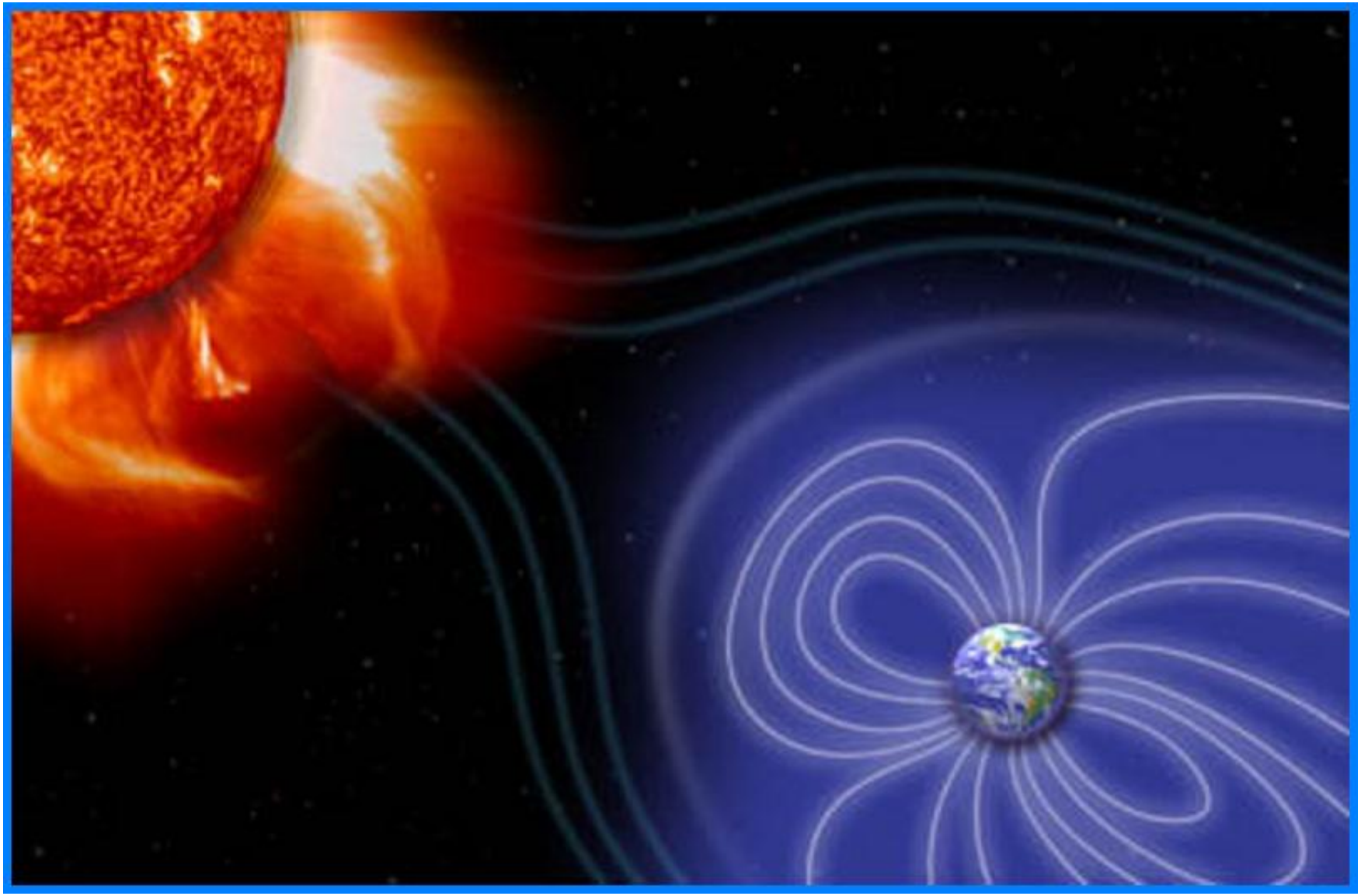


## 北台襲狂 雨雷雹冰

交電雷空上樓大一〇一北台，布密雲烏空天，雨雷大場一了下區地北台大午下天昨集庫水翠翡在下未並雨大場這惜可。怕害人令聲咚咚叮叮，雹冰下降並區地分部，加影攝／原彬高者記

A5見聞新

。大不助幫象旱區地部北解紓對，區水



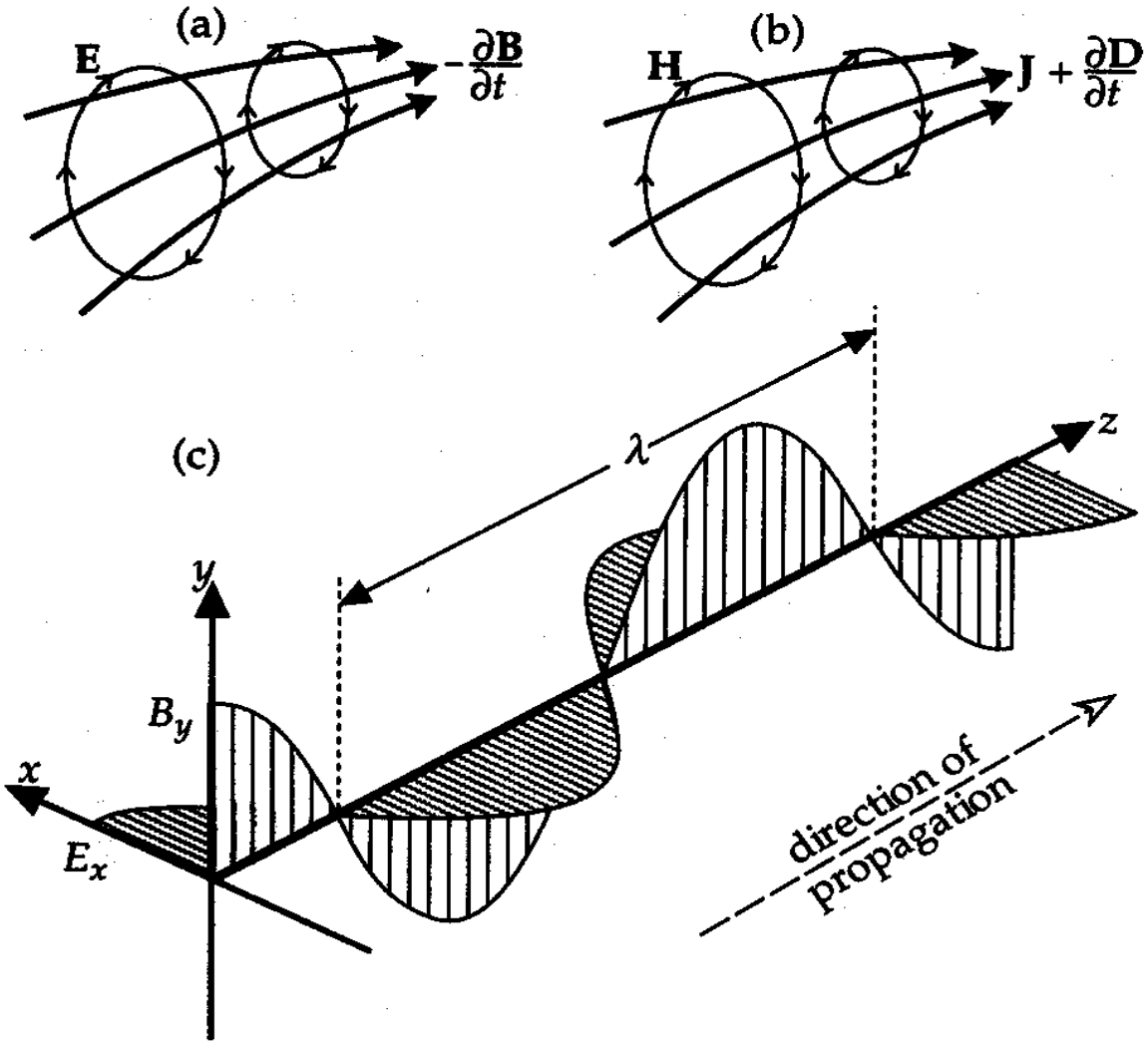
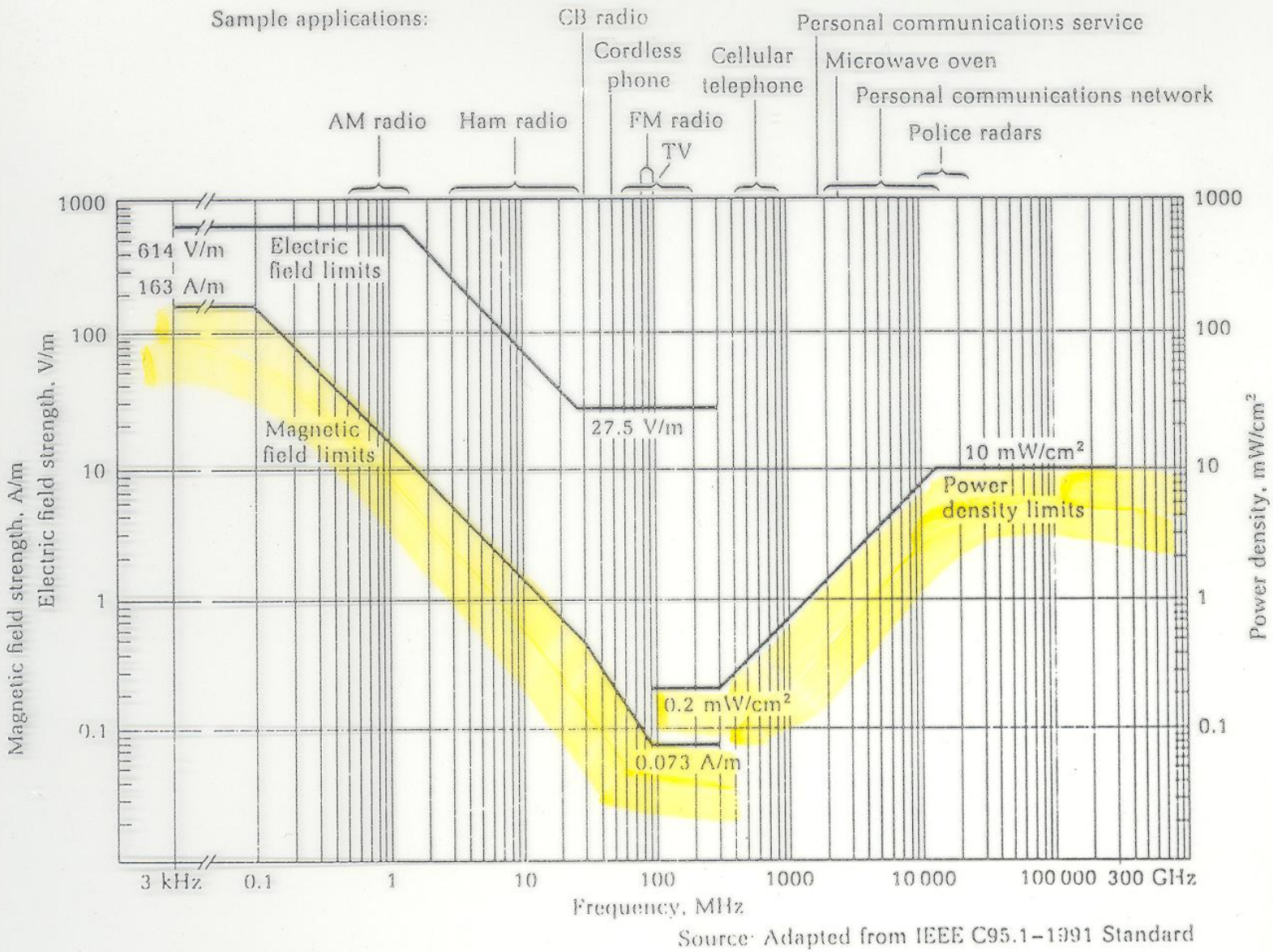


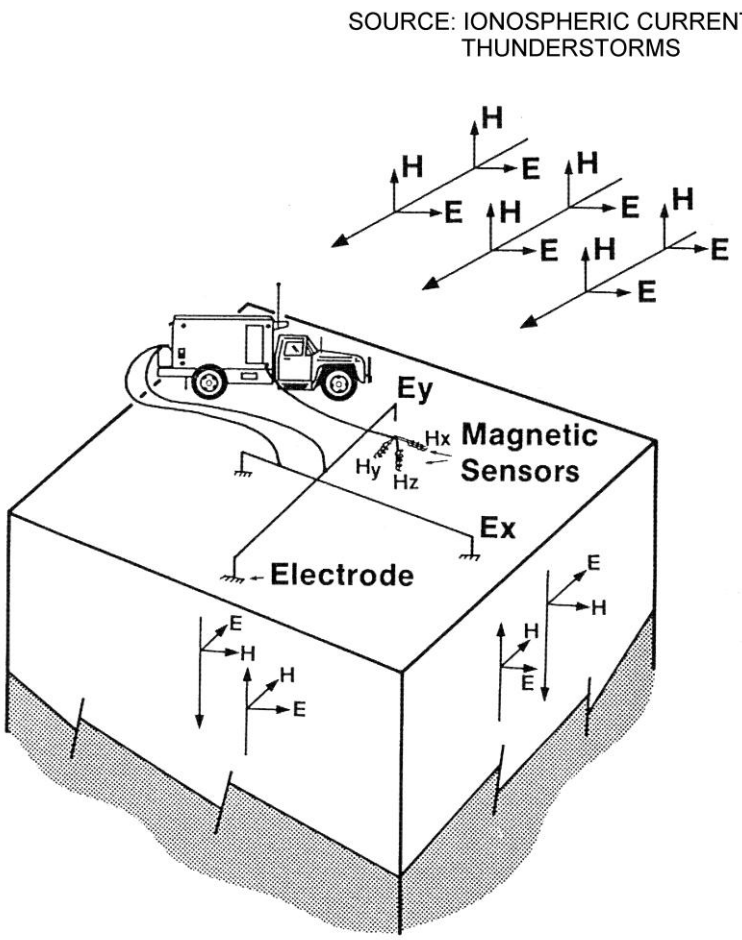

Fig. 4.52 (a) An electric field **E** is generated by a changing magnetic field ($\partial\mathbf{B}/\partial t$), while (b) a magnetic field **B** is produced by the current density **J** and the changing displacement-current density ($\partial\mathbf{D}/\partial t$); (c) in an electromagnetic wave an electric field $E_x$ and a magnetic field $B_y$ fluctuate normal to each other in the plane normal to the propagation direction ($z$-axis).

IEEE safety limits for human exposure to RF electromagnetic fields.




Source: Adapted from IEEE C95.1–1991 Standard

# MT method


SOURCE: IONOSPHERIC CURRENTS
THUNDERSTORMS
H
E
Ey
Hx
Hy
Hz
Magnetic
Sensors
Ex
Electrode

# MT signals….

High frequencies: (>1 Hz)
World-wide lightning discharges



Long periods: (>1 s)
Sunspot activity
Ionosphere-magnetosphere-solar wind-IPMF interactions



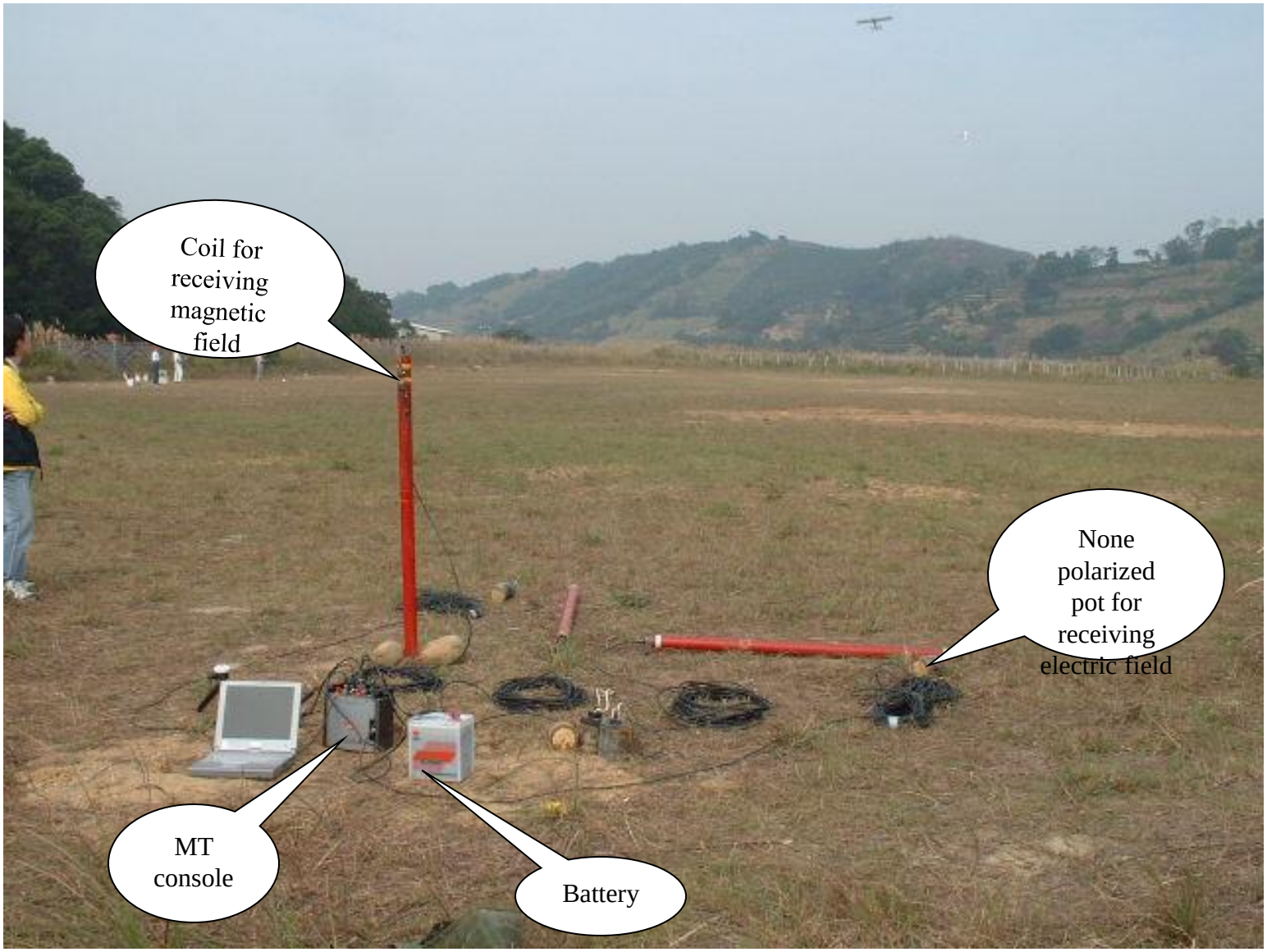
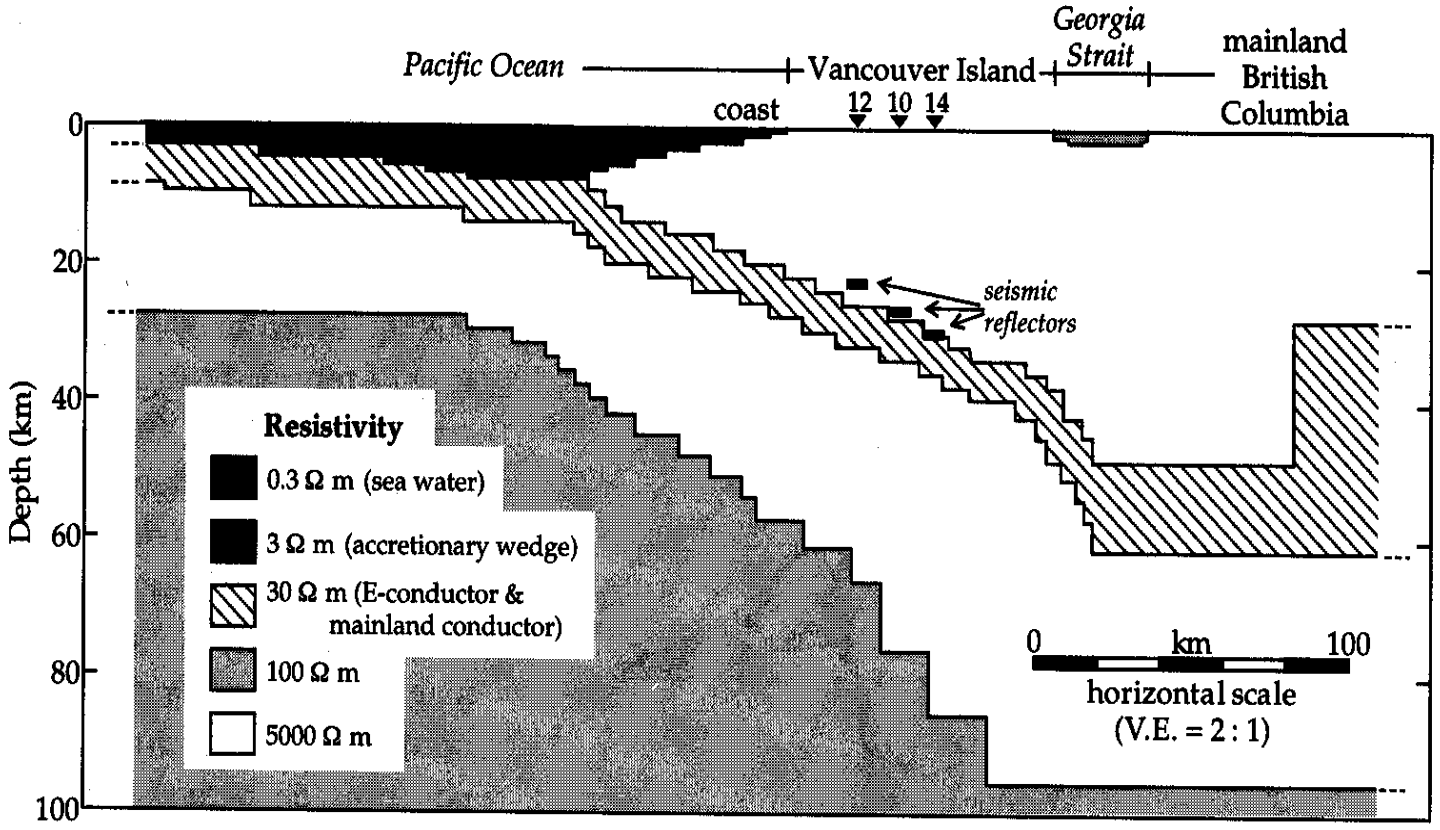
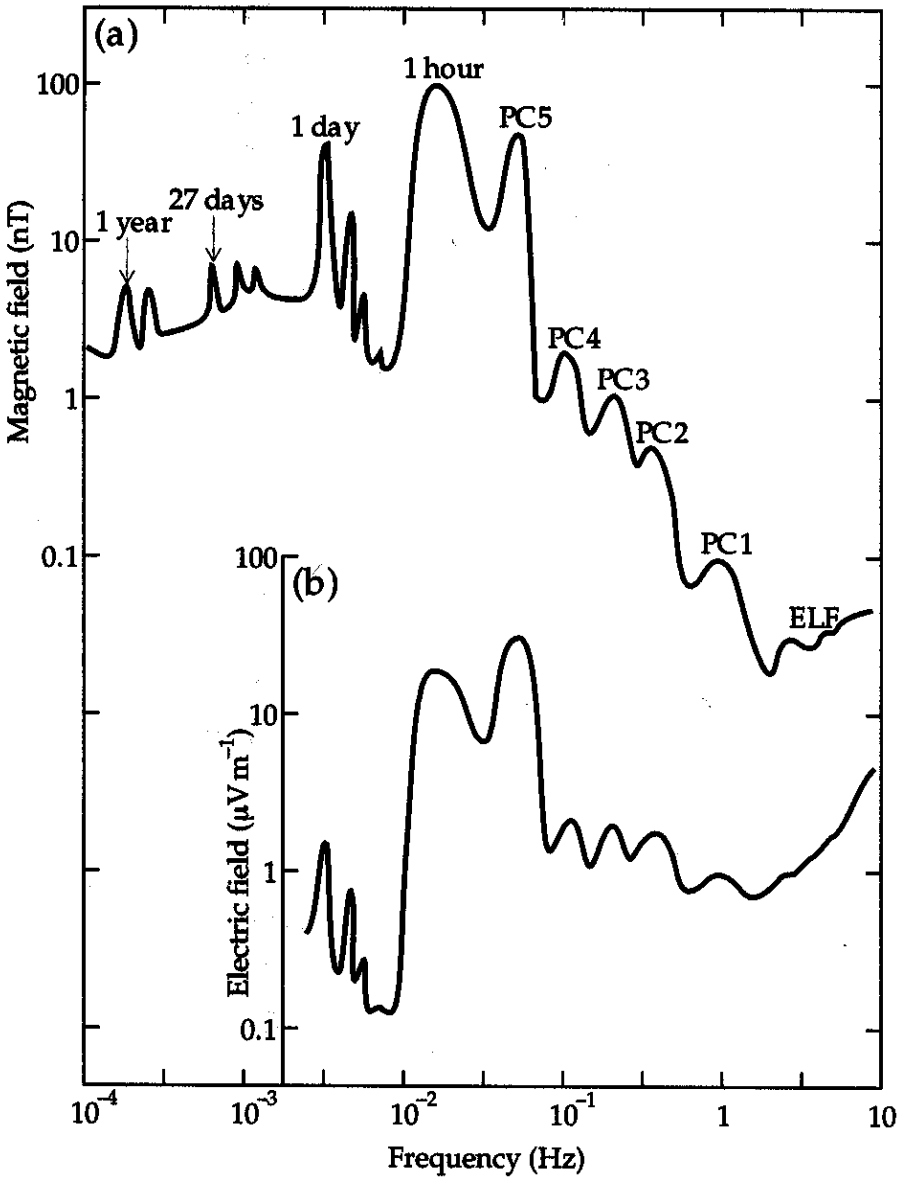

Coil for receiving magnetic field
None polarized pot for receiving electric field
MT console
Battery

Fig. 4.57 Two-dimensional resistivity model of the crust and upper mantle beneath Vancouver Island and the adjacent mainland derived from magnetotelluric results (redrawn from Kurtz *et al.*, 1986).





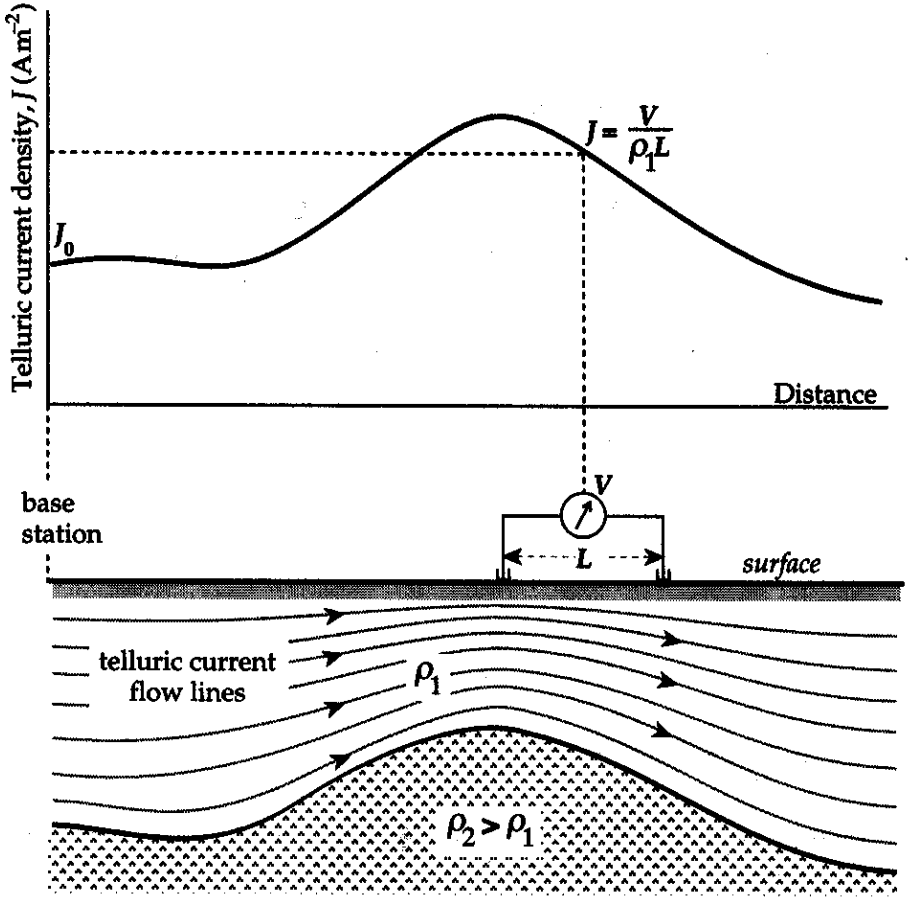
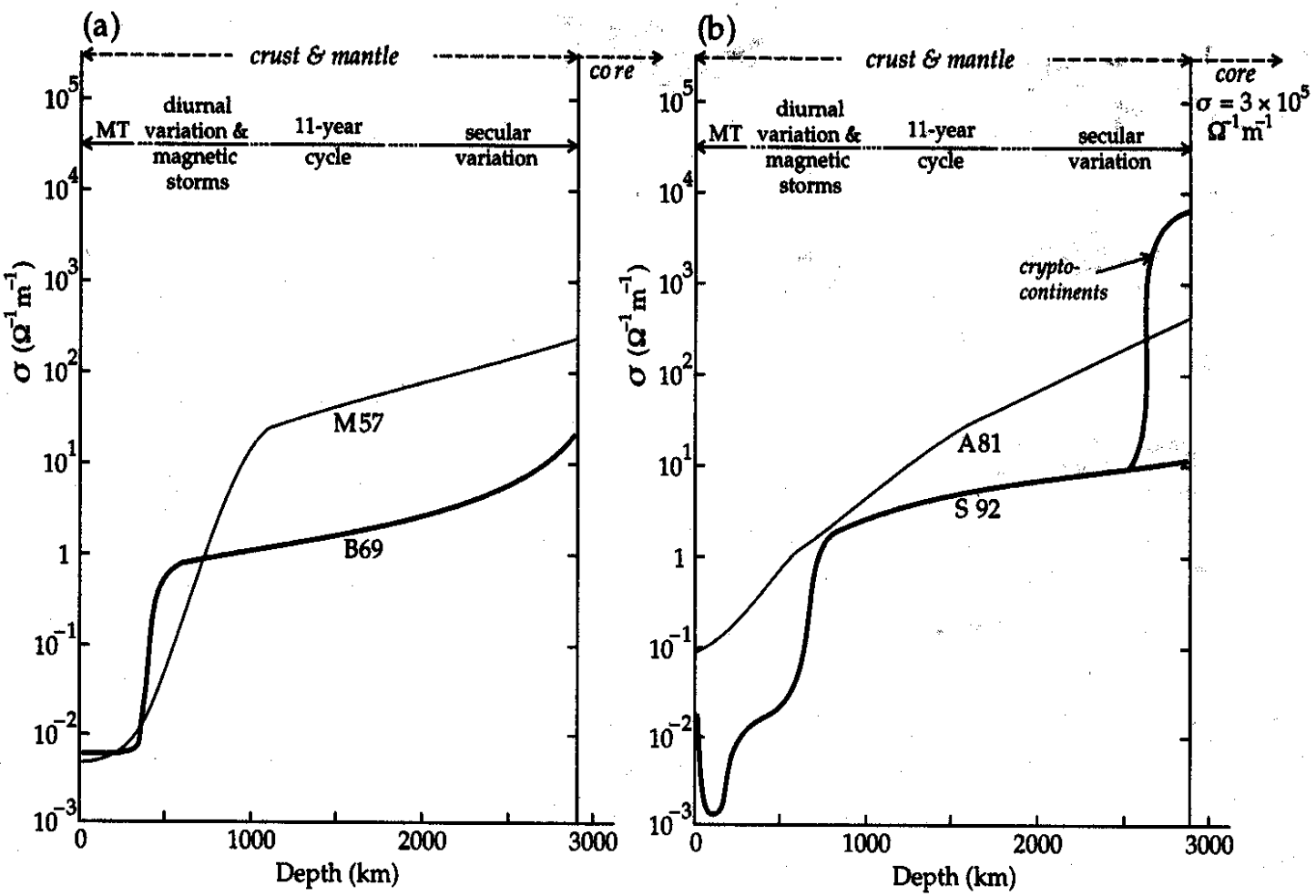




Fig. 4.41 Telluric current lines are deflected by changes in thickness of a conducting layer over a more resistive structure (*bottom*). The telluric current density (*top*) is obtained from the voltage measured between a pair of fixed-separation electrodes at the surface (after Robinson and Çoruh, 1988).

Fig. 4.58 Models of electrical conductivity ($\sigma$) at depth in the mantle proposed by (a) MacDonald (1957; M57) and Banks (1969; B69), (b) Achache *et al.*, (1981; A81) and Stacey (1992; S92).



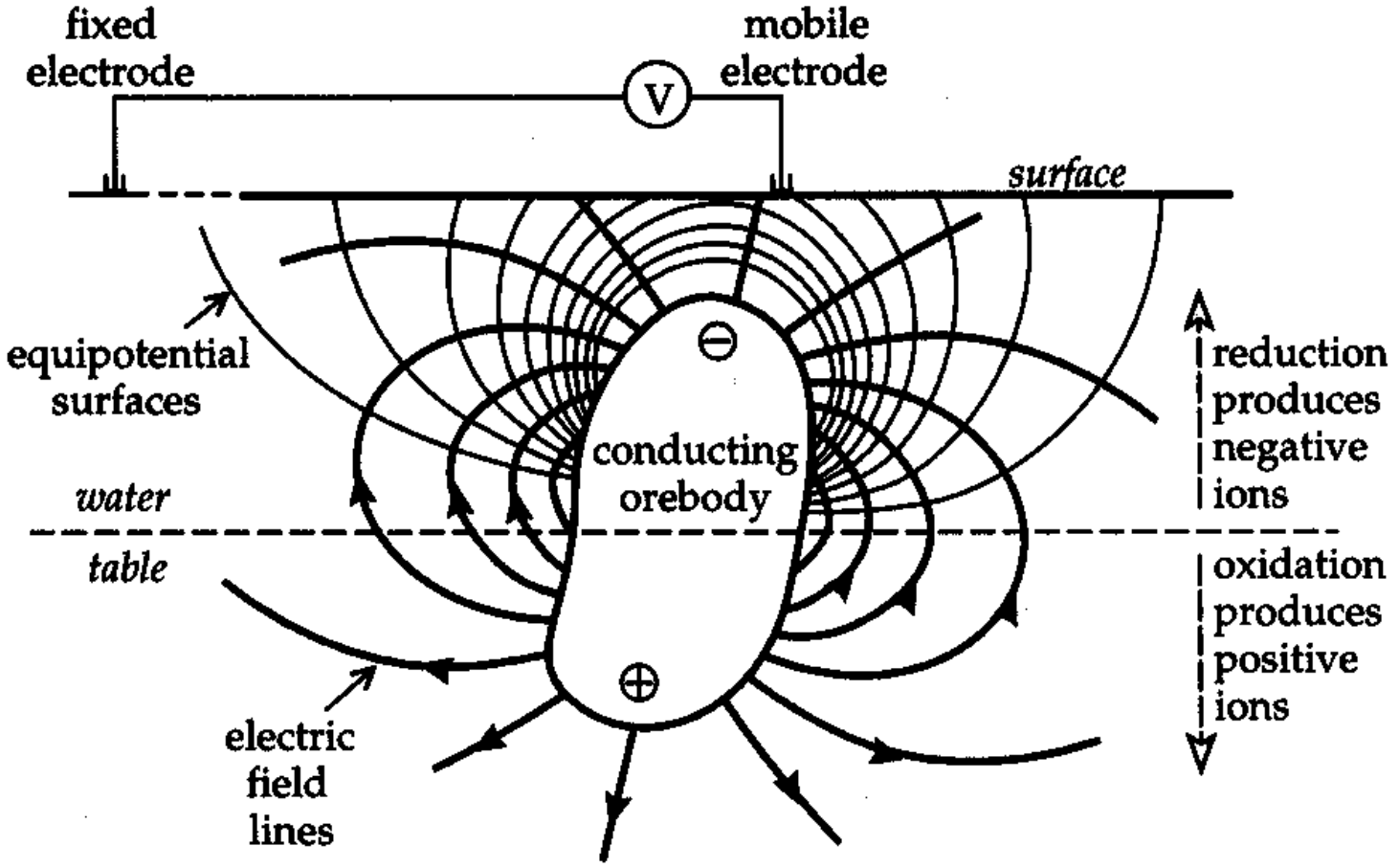



Fig. 4.37 A schematic model of the origin of the self-potential anomaly of an orebody. The mechanism depends on differences in oxidation potential above and below the water table.

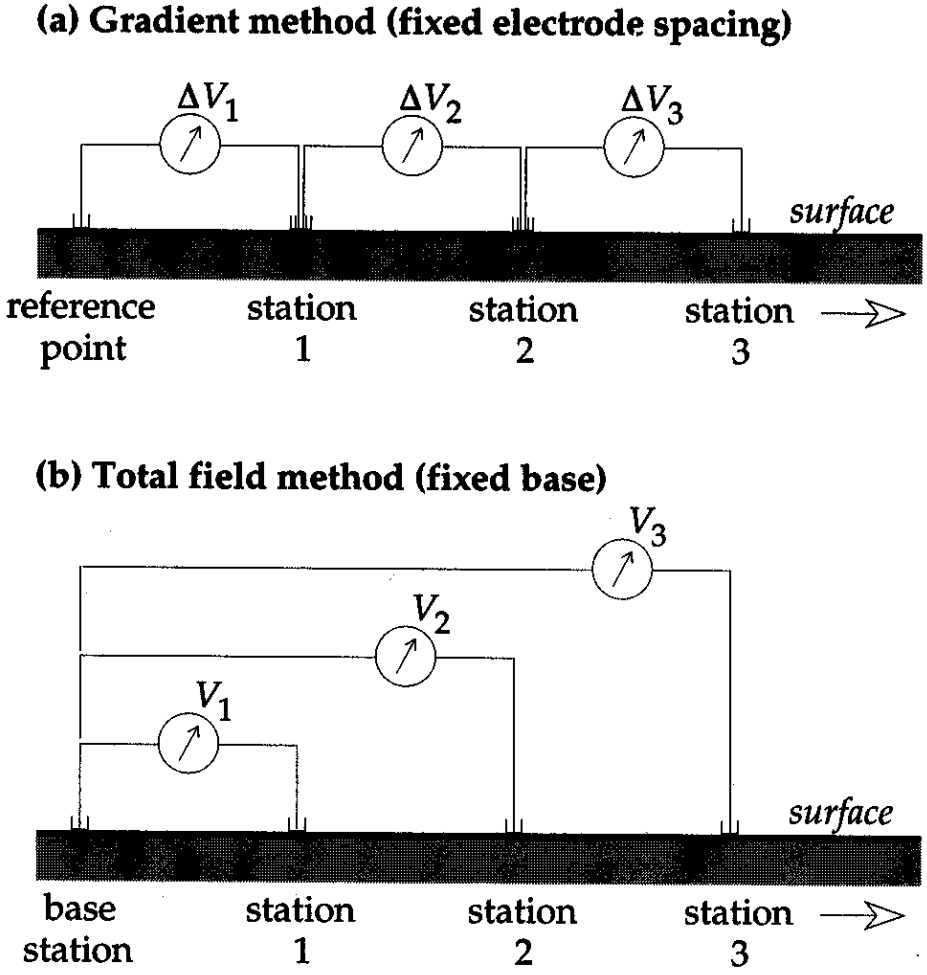



Fig. 4.38 The field techniques of measuring self-potential by (a) the gradient method and (b) the total field method. The total potential $V$ at a station in the gradient method is found by summing the previous potential differences $\Delta V$; in the total field method $V$ is measured directly.

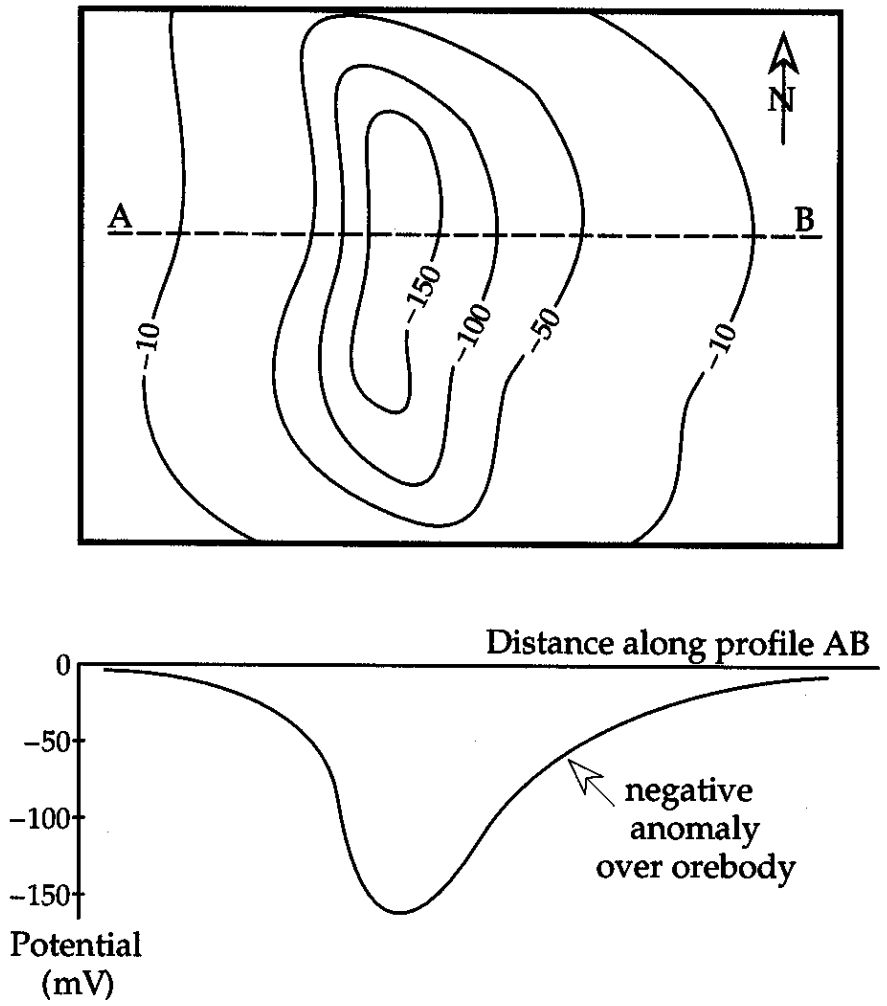
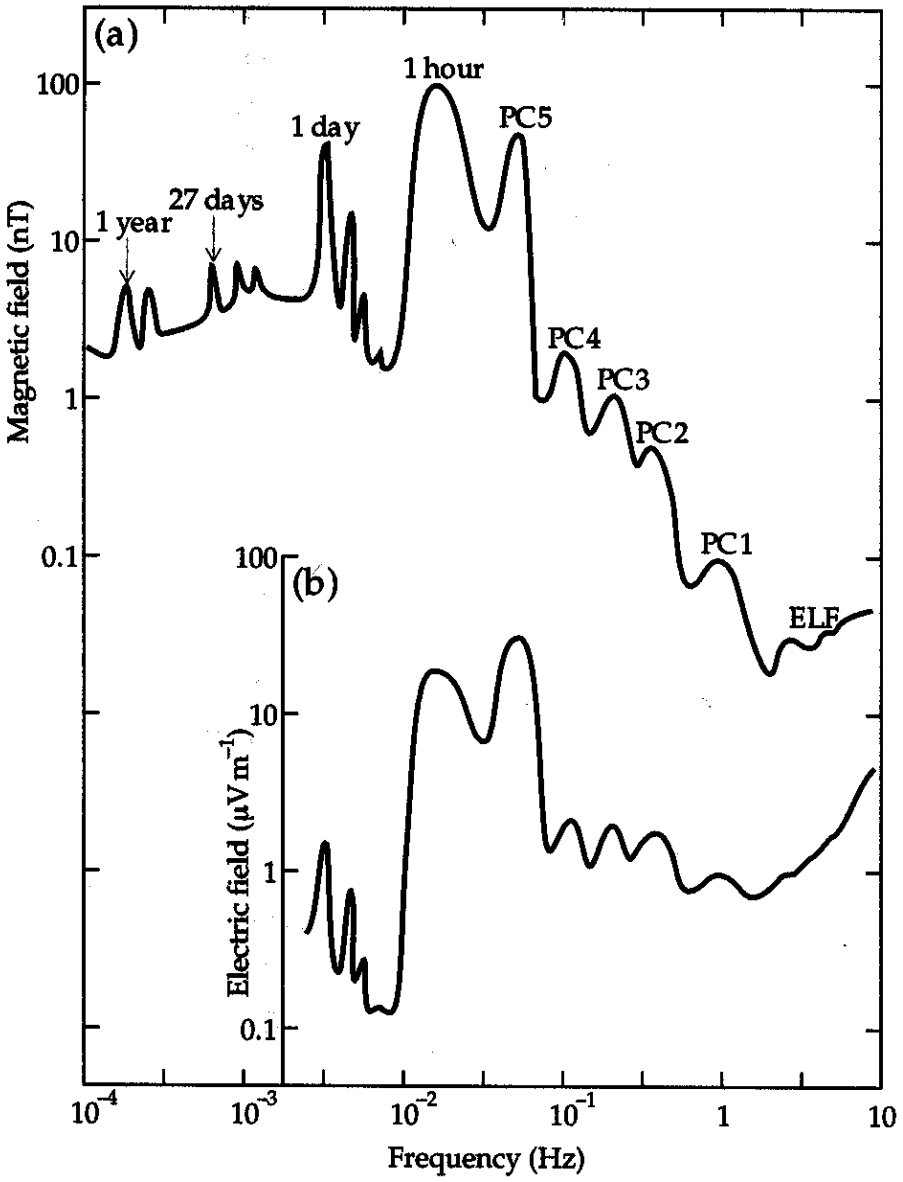



Fig. 4.39 Hypothetical contour lines of a negative self-potential anomaly over an orebody; the asymmetry of the anomaly along the profile AB suggests that the orebody dips toward A.



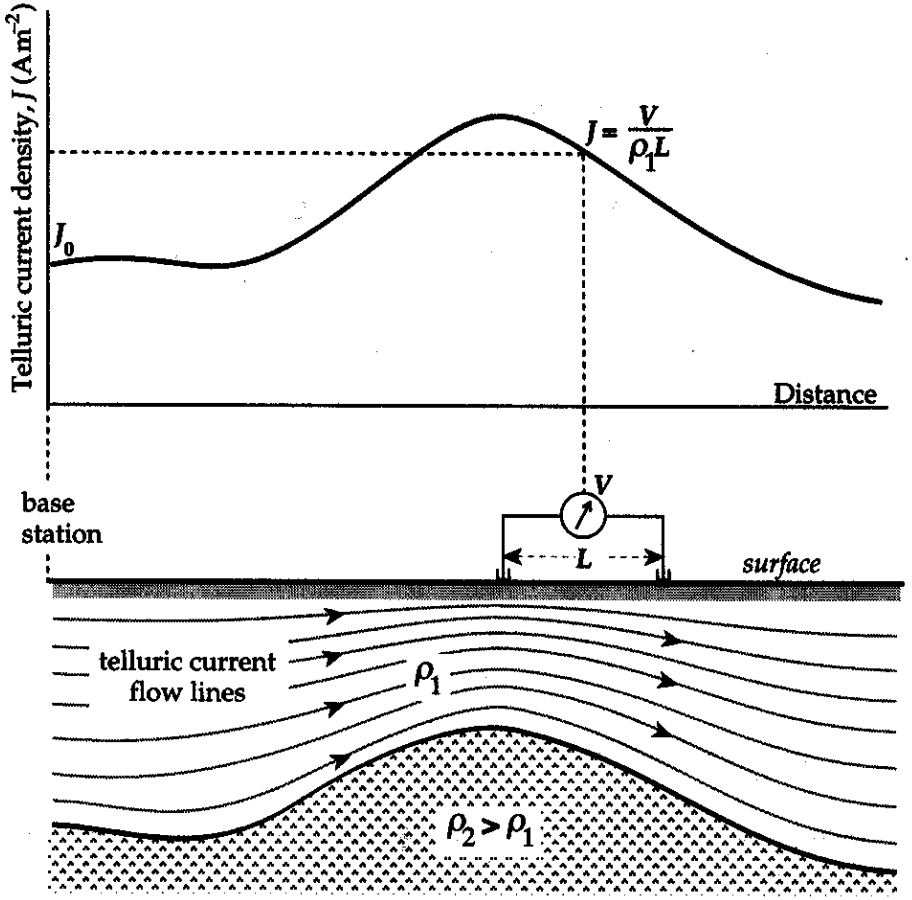




Fig. 4.41 Telluric current lines are deflected by changes in thickness of a conducting layer over a more resistive structure (*bottom*). The telluric current density (*top*) is obtained from the voltage measured between a pair of fixed-separation electrodes at the surface (after Robinson and Çoruh, 1988).

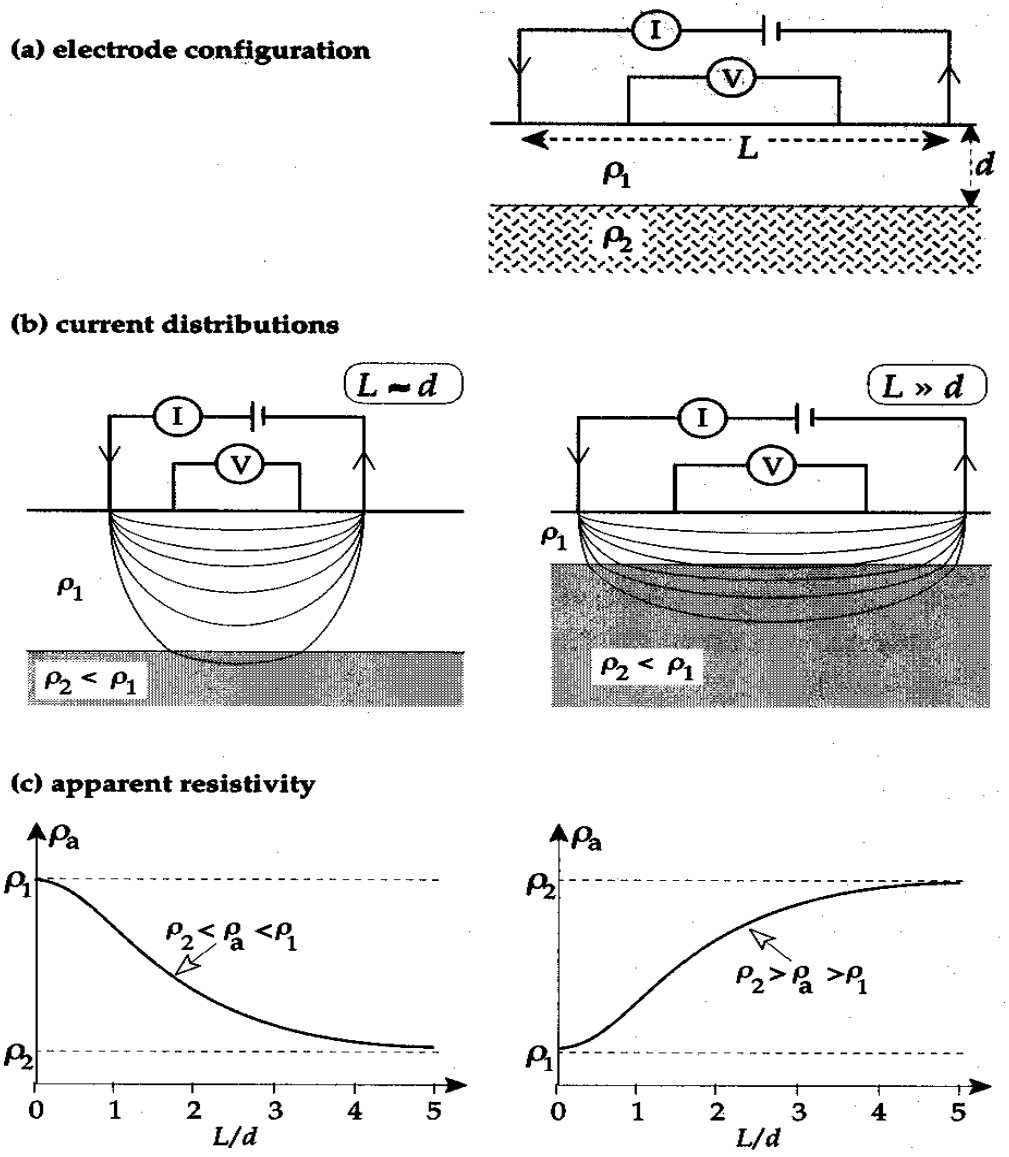



Fig. 4.47 (a) Parameters of the four-electrode arrangement, (b) distribution of current lines in a two-layer ground with resistivities $\rho_1$ and $\rho_2$ ($\rho_1 > \rho_2$) and (c) the variation of apparent resistivity as the current electrode spacing is varied for the two cases of $\rho_1 > \rho_2$ and $\rho_1 < \rho_2$.

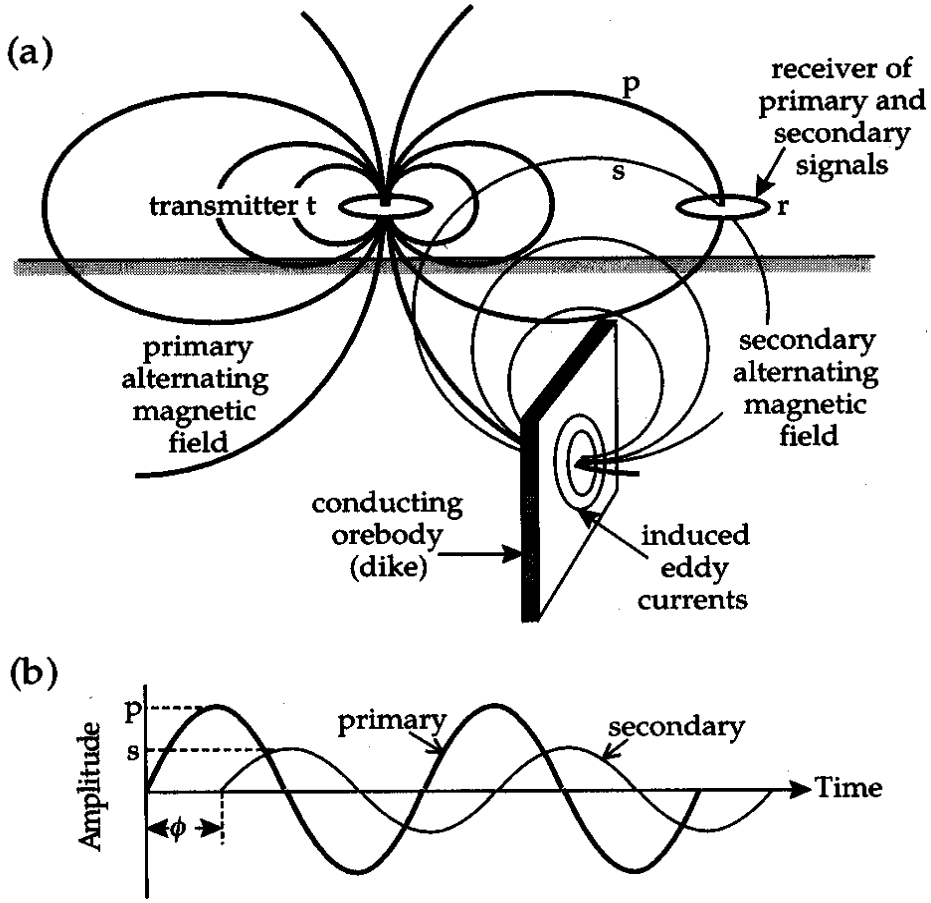



Fig. 4.55 (a) Illustration of primary and secondary fields in the horizontal loop induction method of electromagnetic exploration for shallow orebodies. (b) Amplitudes and phases of the primary (p) and secondary (s) fields.

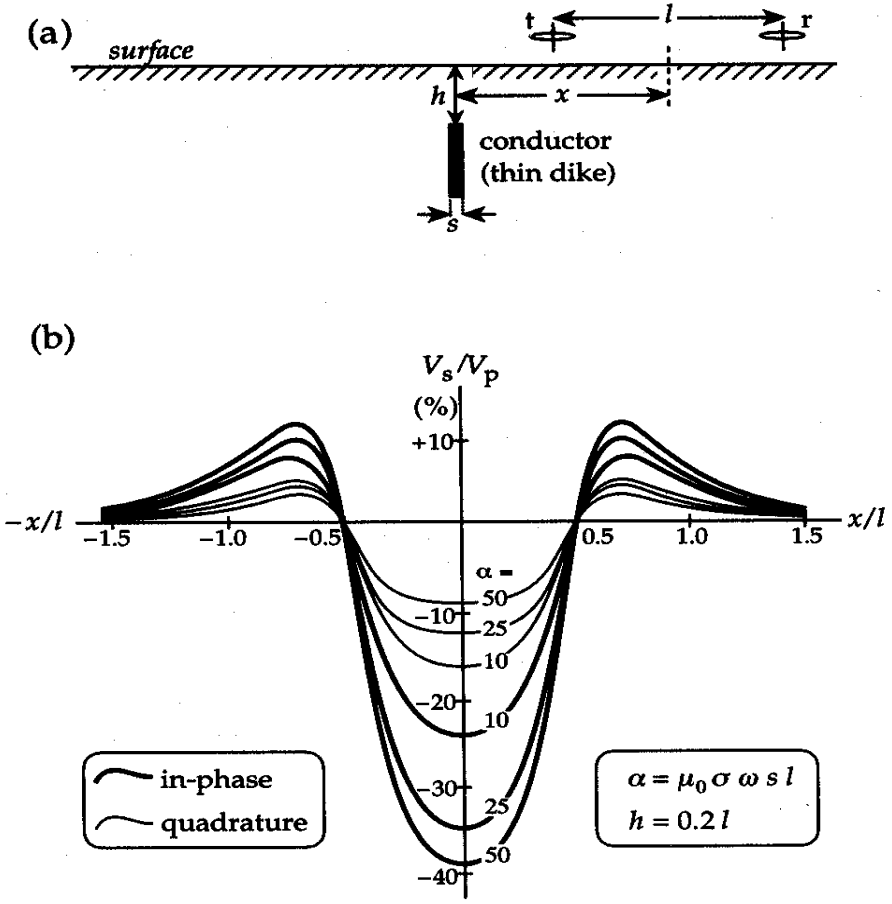



Fig. 4.56 (a) Geometry of an HLEM profile across a thin vertical dike. (b) In-phase and quadrature profiles over a dike at depth $h/l = 0.2$ for some values of the response parameter $\alpha$.

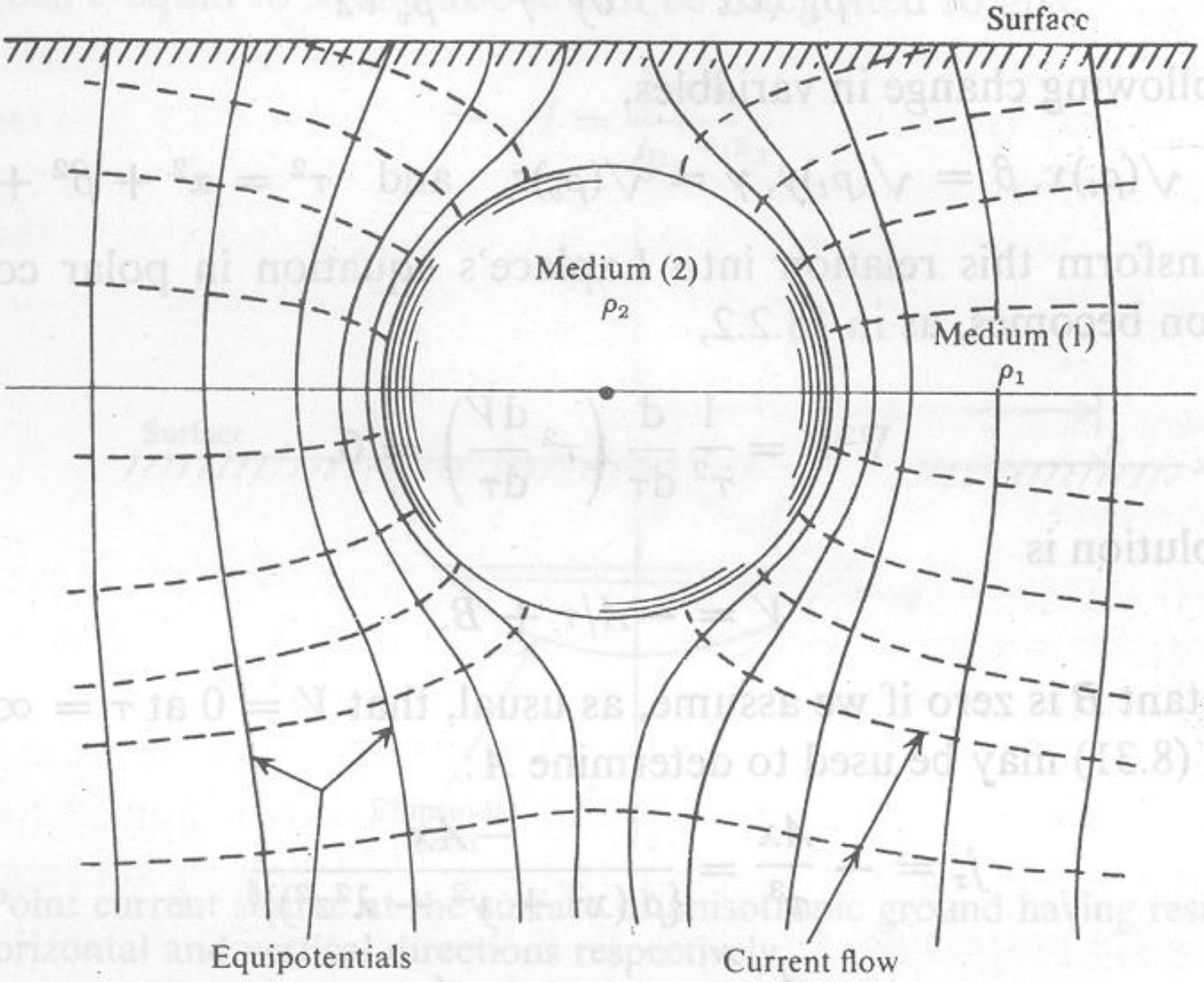



Fig. 8.15 Equipotentials and current flow-lines for buried conductive sphere.

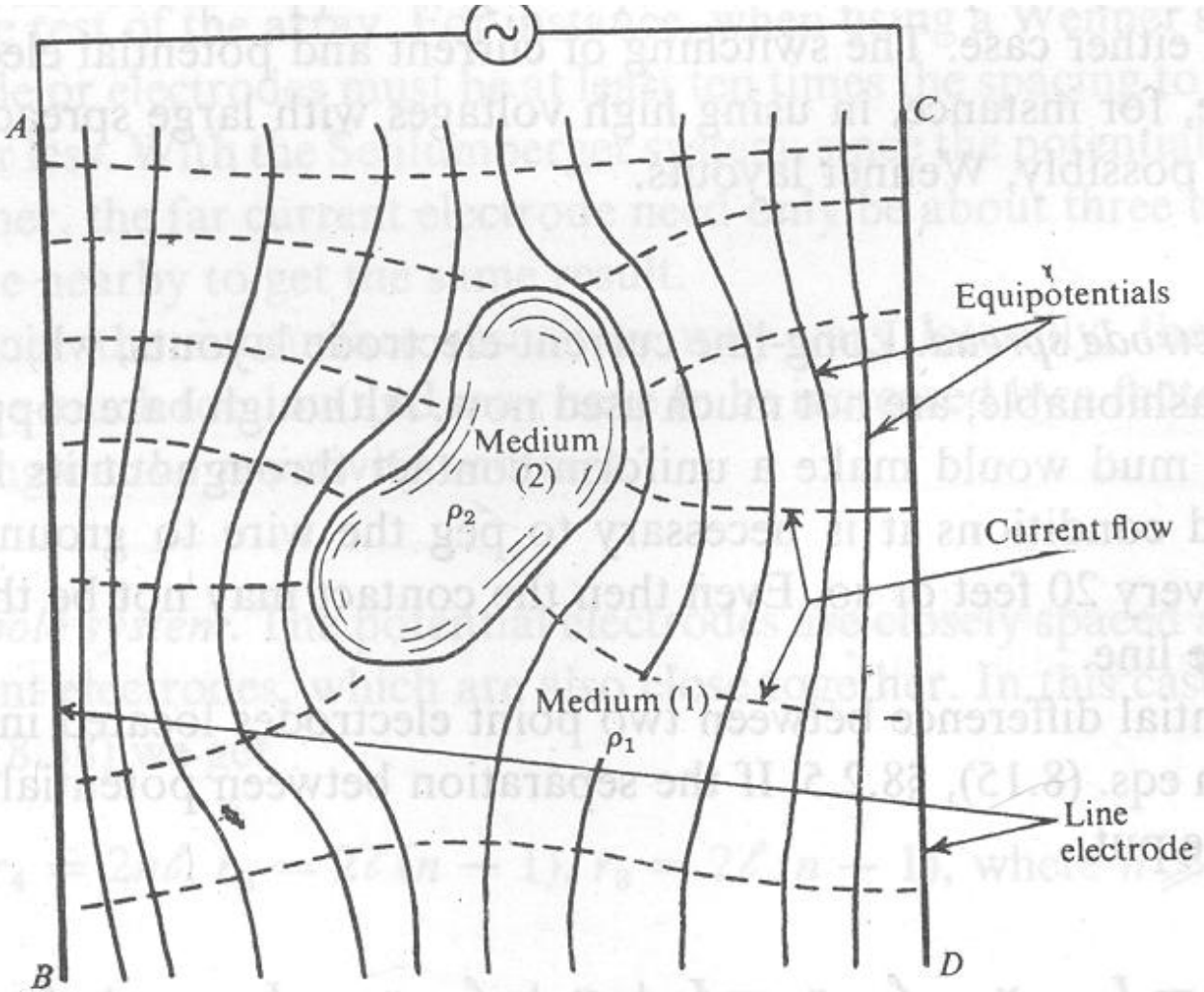



Fig. 8.20 Distortion of equipotentials of a line-electrode system due to the presence of a conductor.

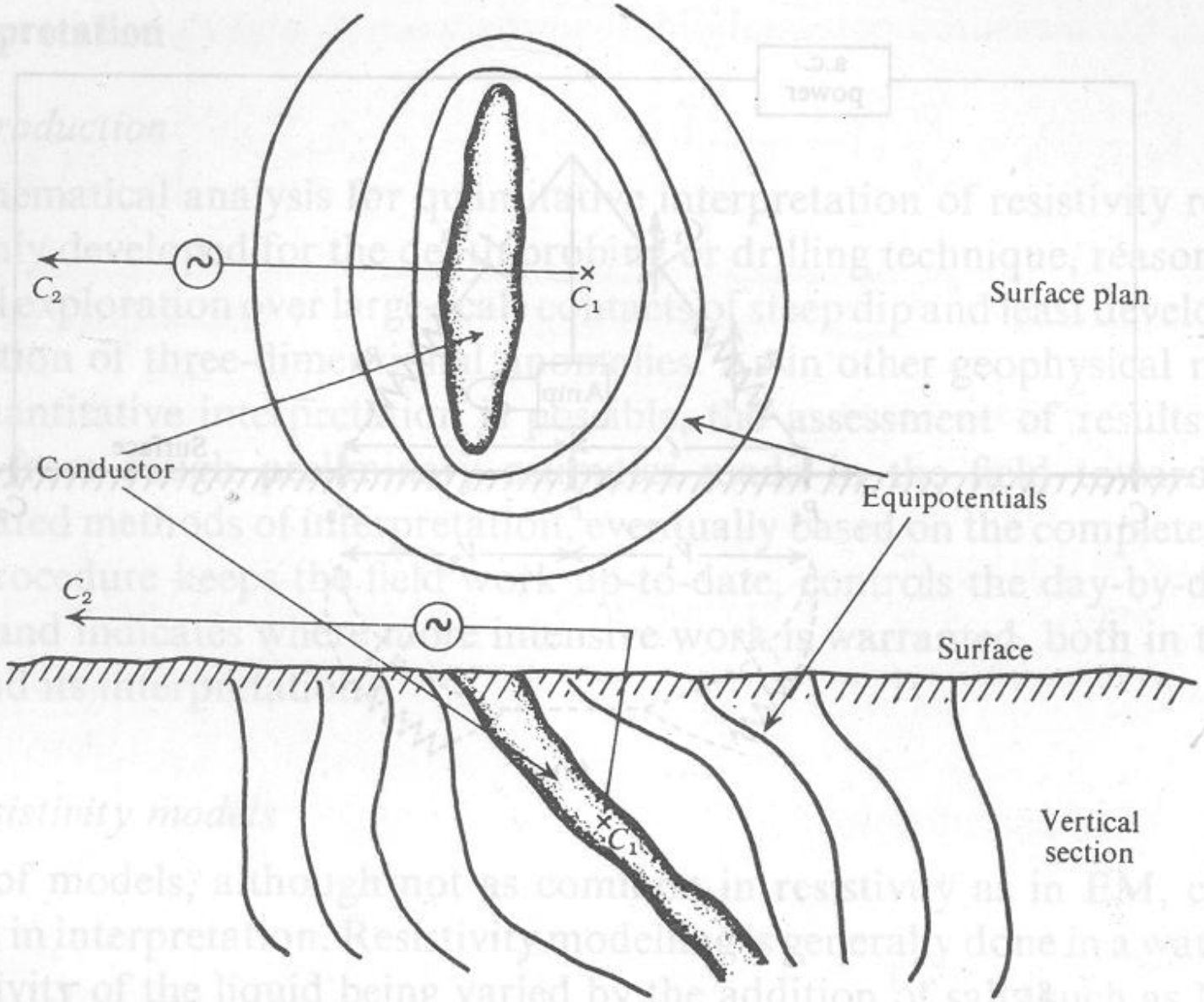



Fig. 8.21 Distortion of the equipotentials around the near electrode by a dipping conductor when using the mise-à-la-masse method.

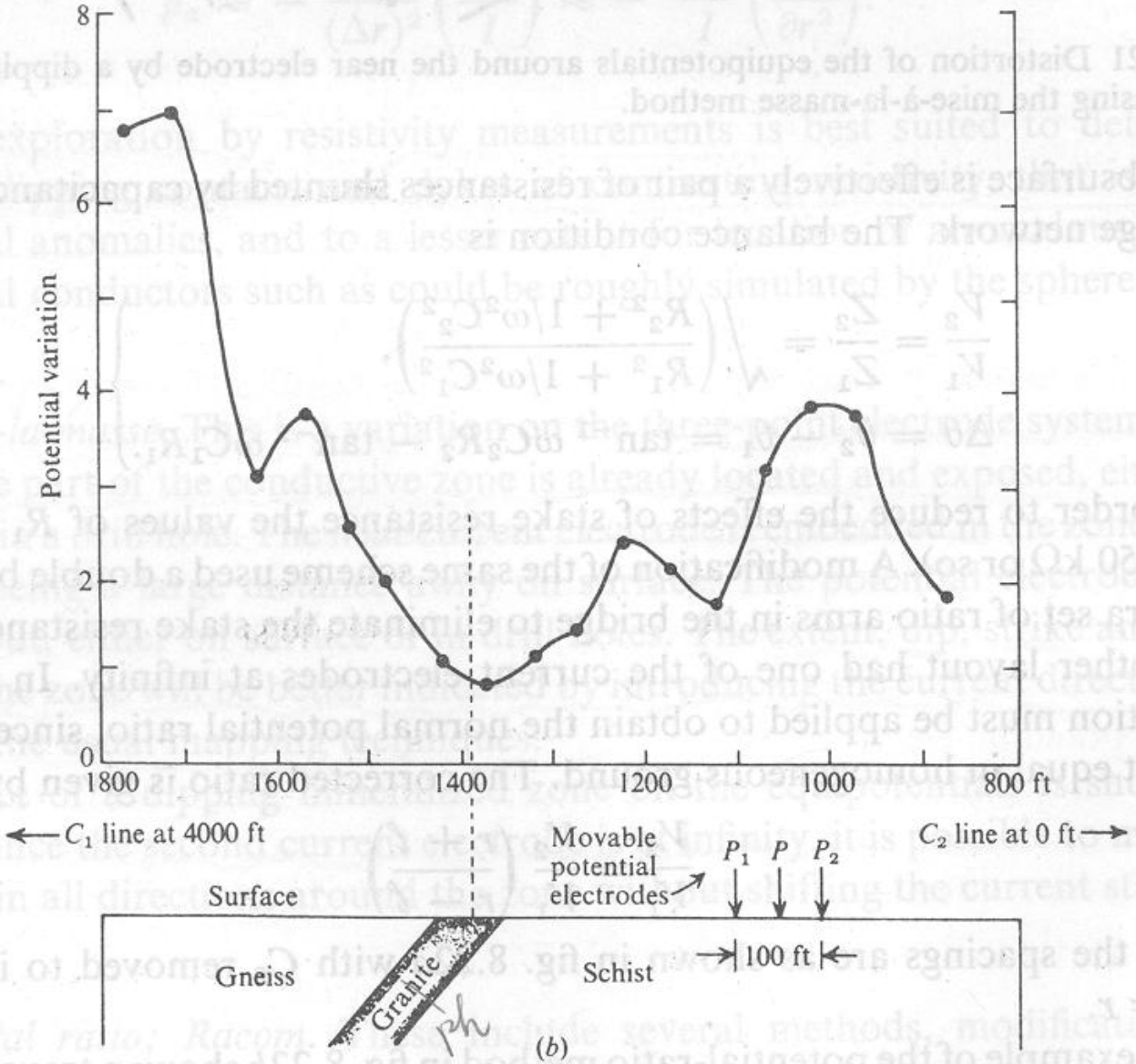



Fig. 8.22 Potential-ratio (Racom) method. (*a*) Field instrument (schematic). (*b*) profile over graphite zone.

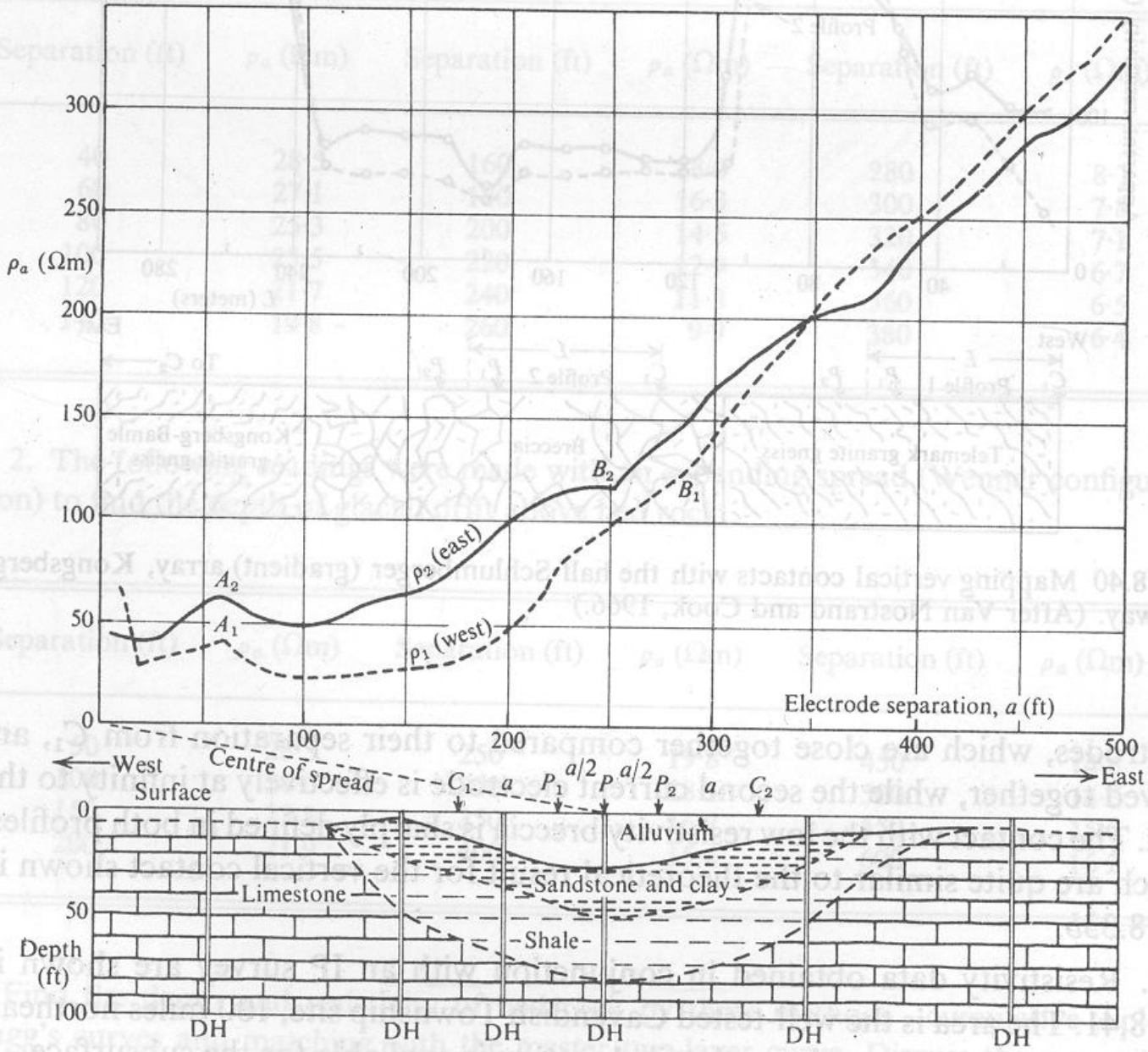



Fig. 8.39 Expanding Lee spread over filled sinkhole, Cherokee County, Kansas. (After Van Nostrand and Cook (1966).)

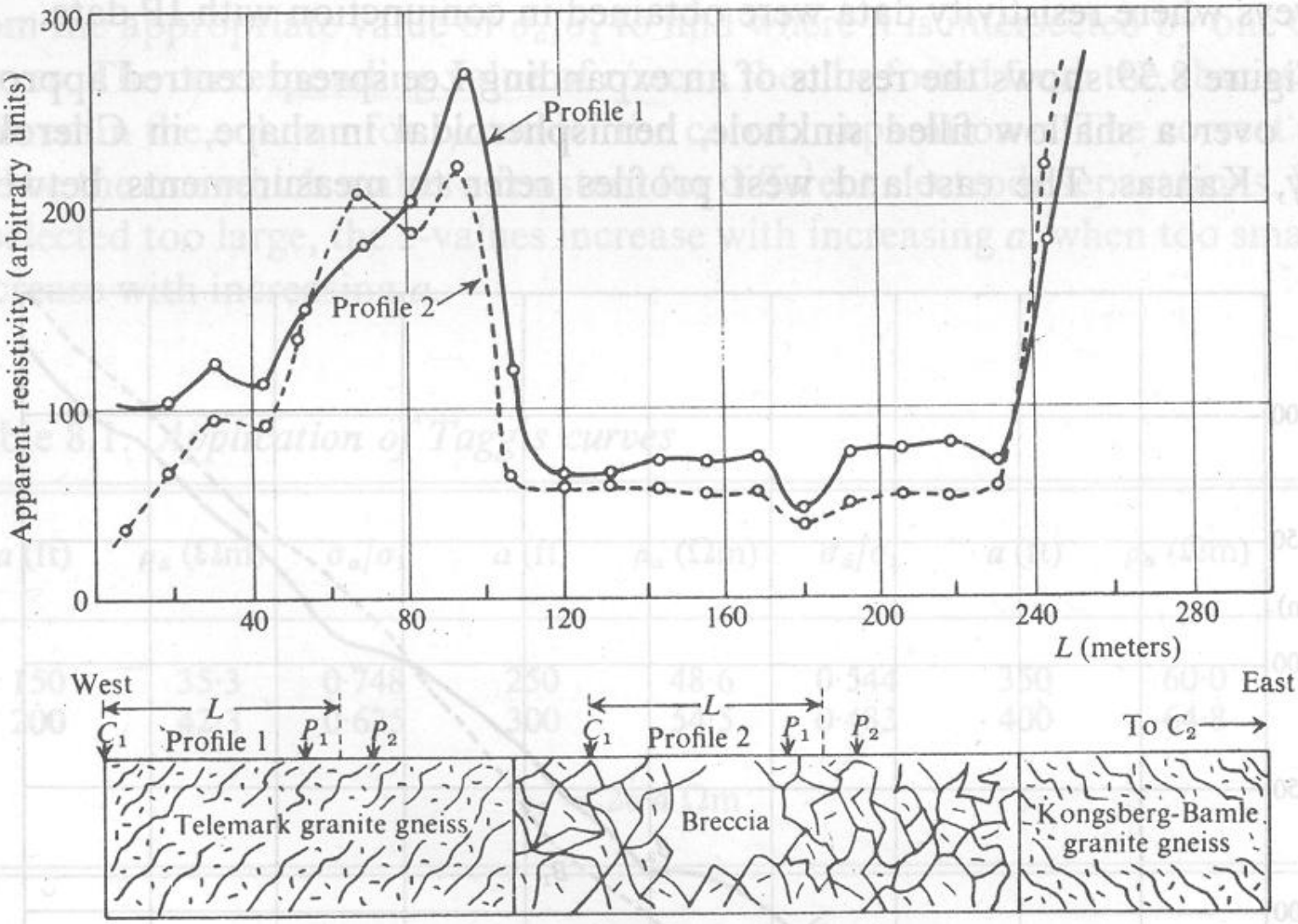



Fig. 8.40 Mapping vertical contacts with the half-Schlumberger (gradient) array, Kongsberg, Norway. (After Van Nostrand and Cook, 1966.)

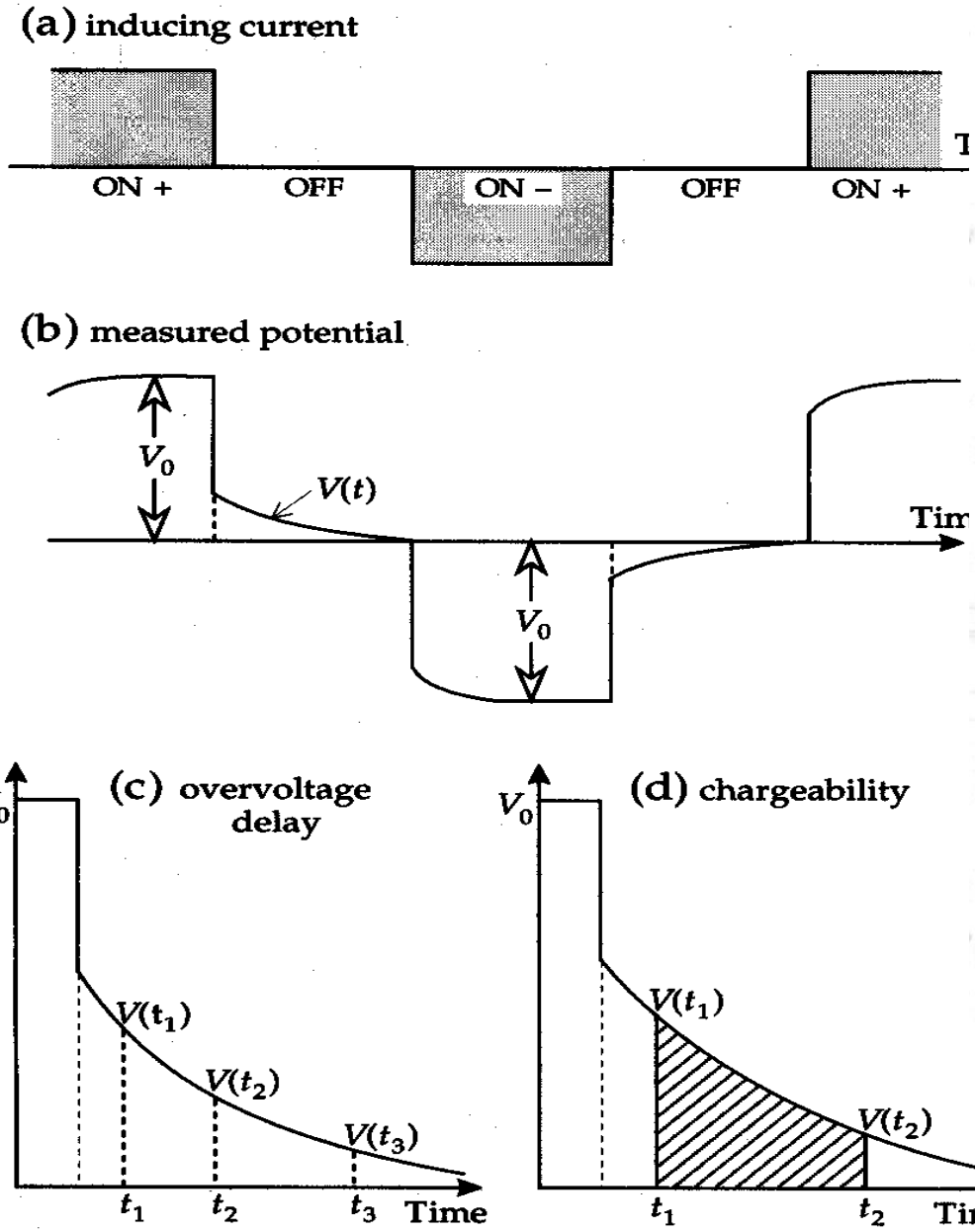



**Fig. 4.50 (a) Illustration of the IP-related decay of potential after interruption of the primary current. (b) Effect of the IP decay time on the potential waveform for a square-wave input current.**

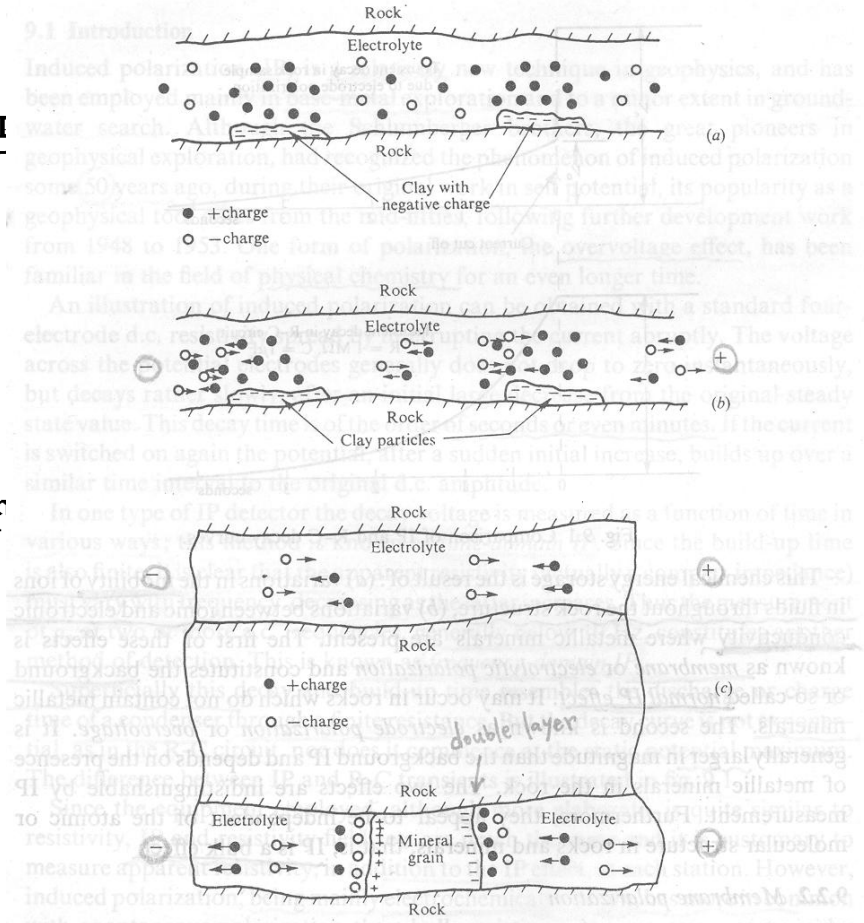



Fig. 9.2 Membrane and electrode polarization effects. (*a*) Normal distribution of ions in a porous sandstone; (*b*) membrane polarization in a porous sandstone due to an applied d.c. voltage; (*c*) electrolytic flow in upper pore, electrode polarization in lower pore.

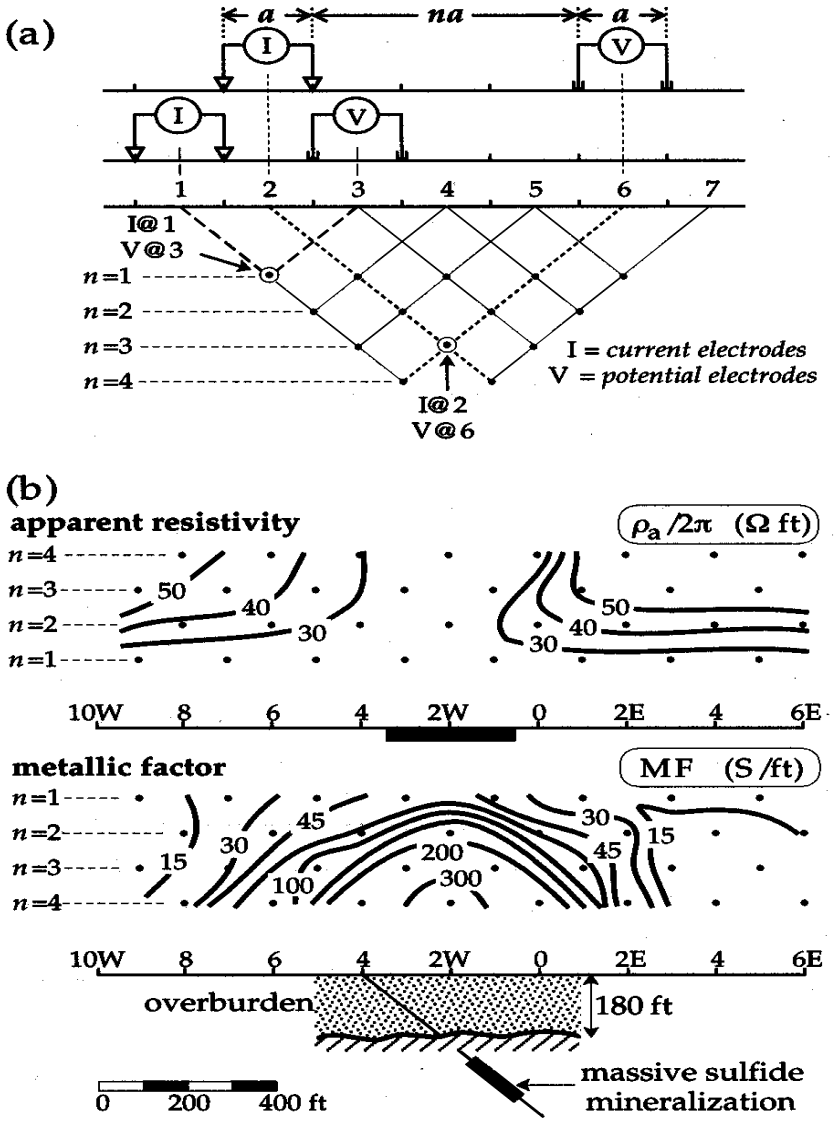



Fig. 4.51 (a) Construction of a pseudo-section for a double-dipole IP survey: the measured parameter is plotted at the intersection of 45° lines extending from the mid-points of the transmitter and receiver pairs. (b) Pseudo-sections of apparent resistivity and metallic factor for an IP survey over a sulfide orebody (redrawn from Telford *et al.*, 1990).

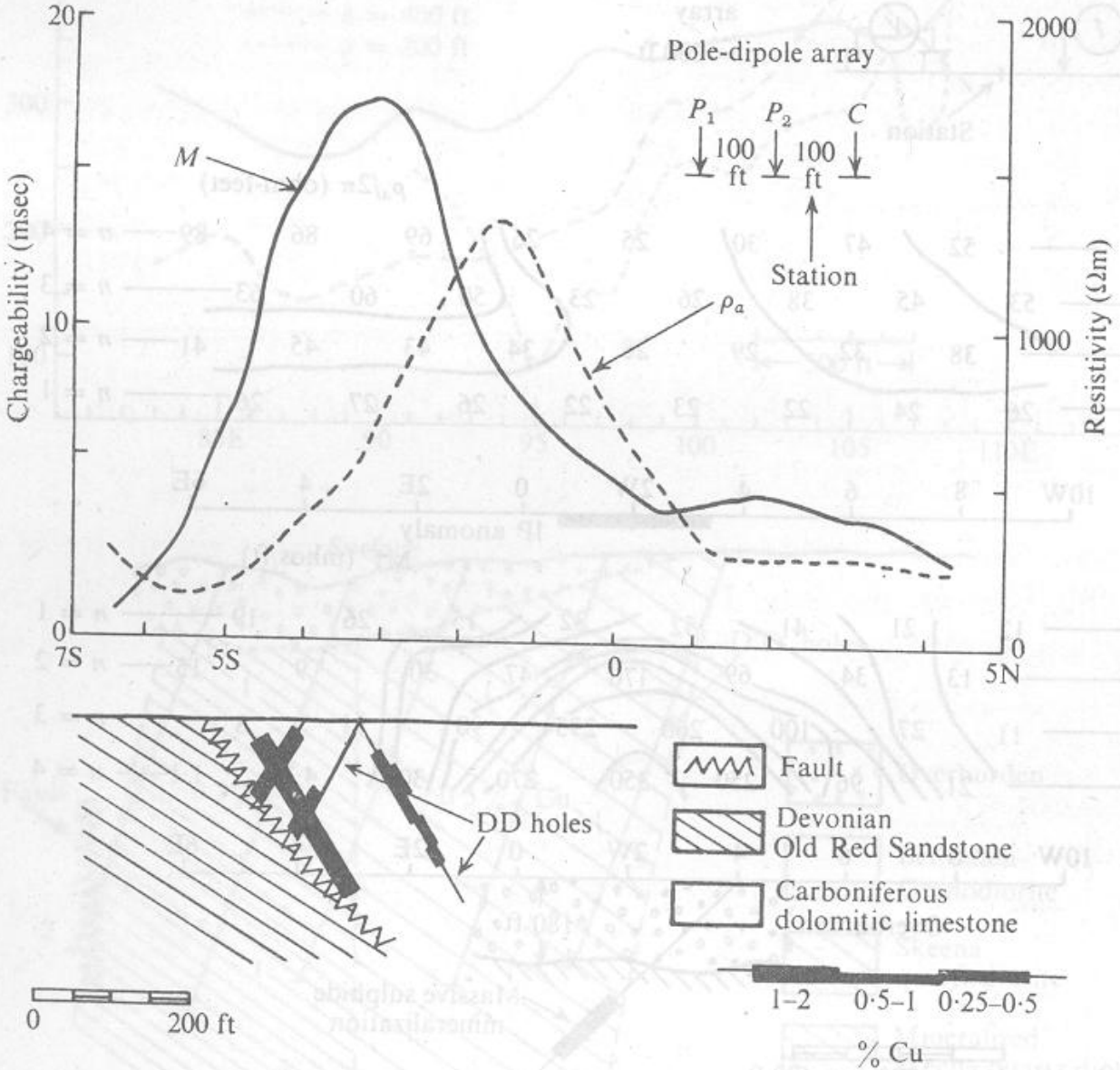



Fig. 9.14 Time-domain IP over Gortdrum copper-silver body. (From Seigel, 1967.)

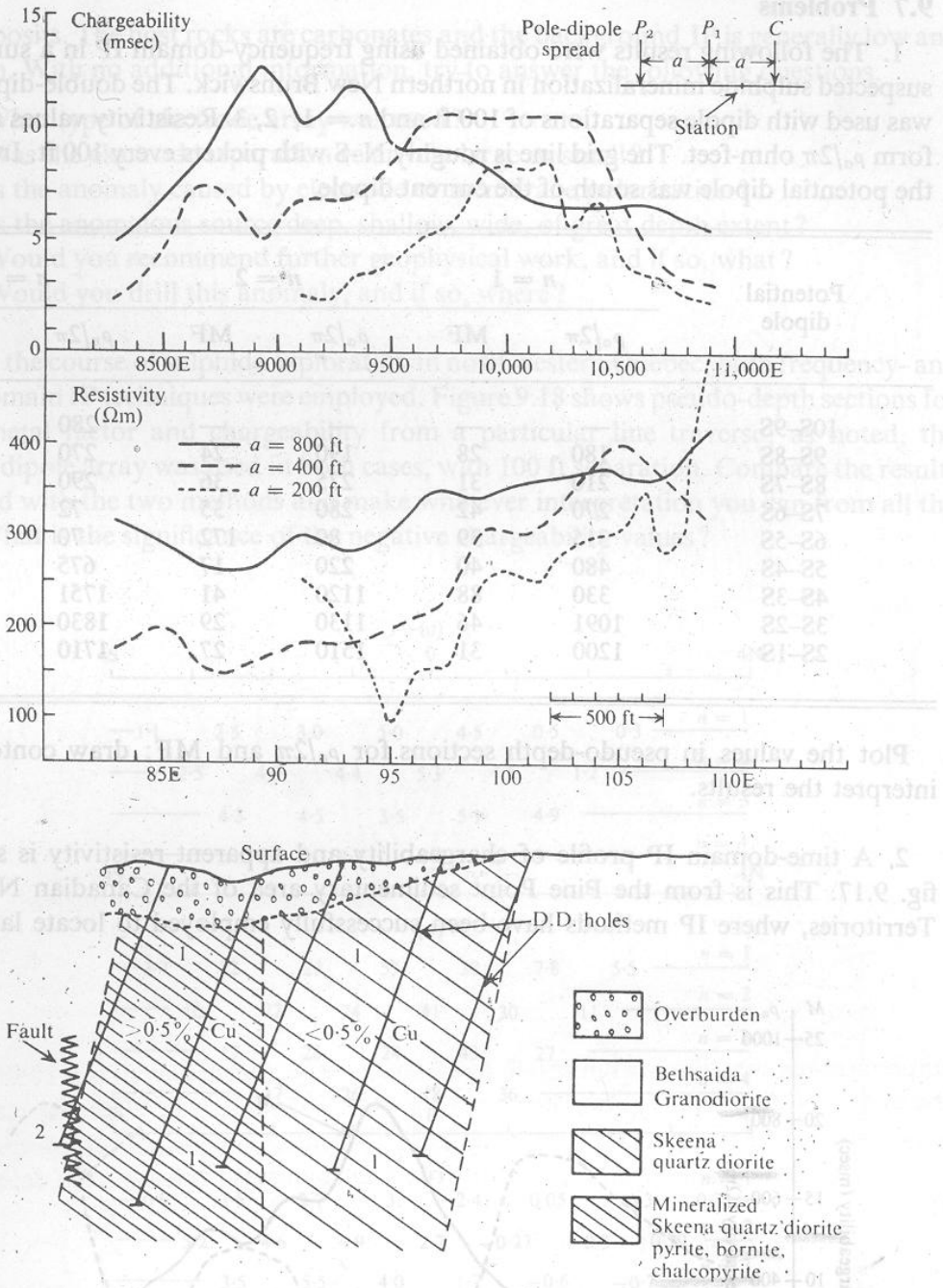



Fig. 9.16 Time-domain IP results over porphyry-copper deposit. (After Seigel, 1967.)

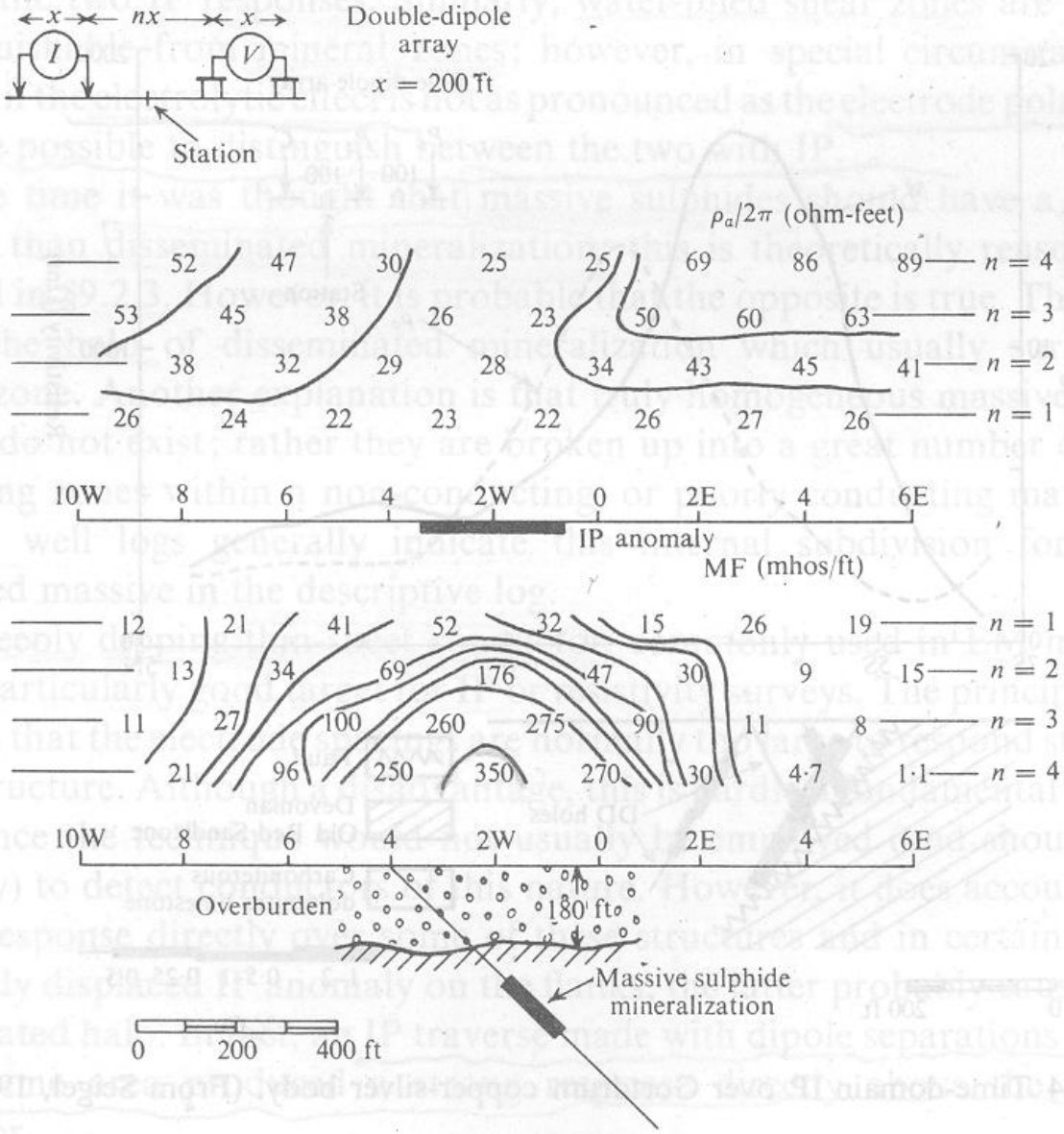
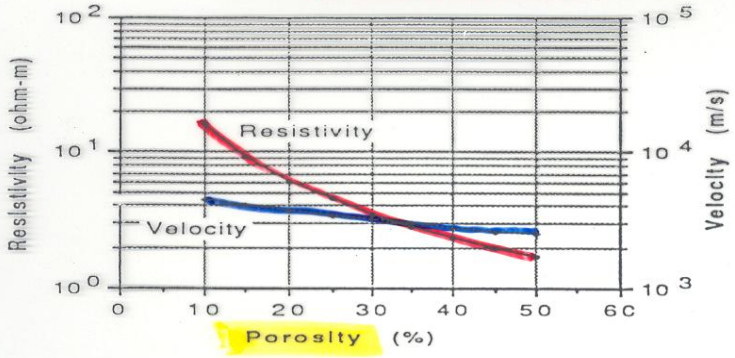
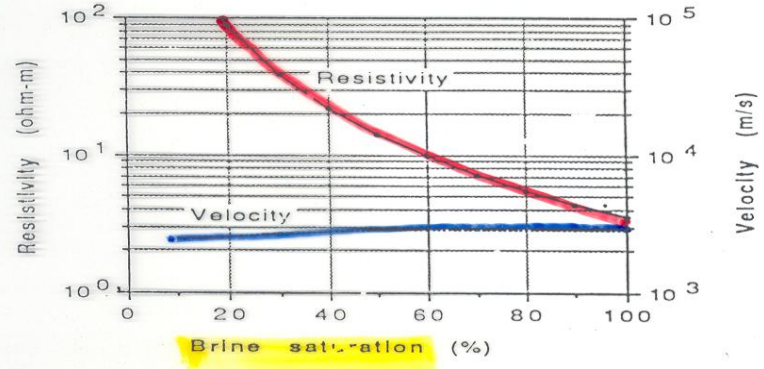
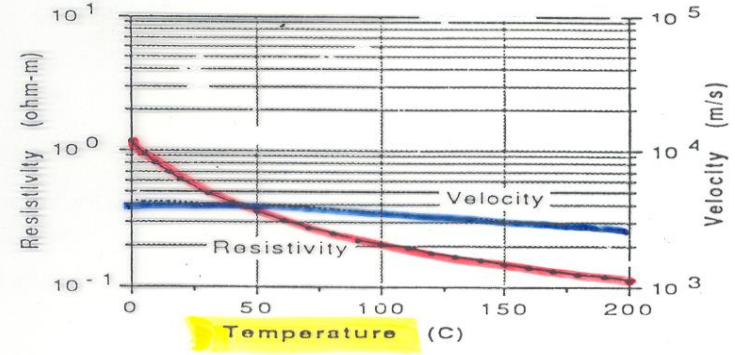



Fig. 9.15 Frequency-domain IP results for massive sulphides overlain by thick conductive overburden.

# Resistivity and velocity as a function of porosity, $\Phi$, water saturation, $S_w$, and temperature, °C.



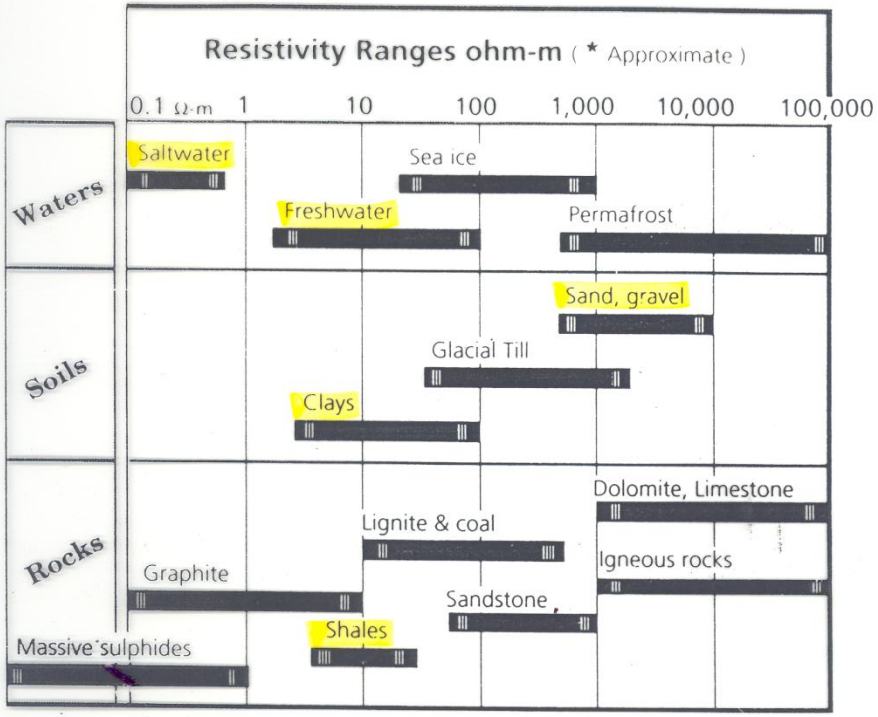

Unconsolidated sandstone

$\rho_w$ = 0.65 ohm-m at 20 °C

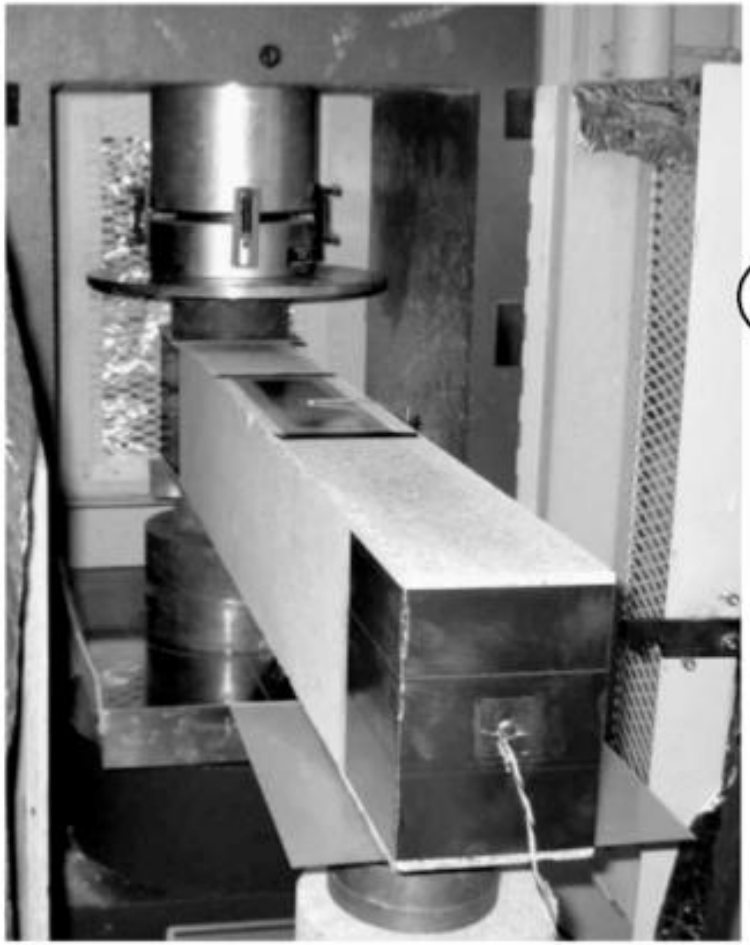
$\Phi$ = 30 %

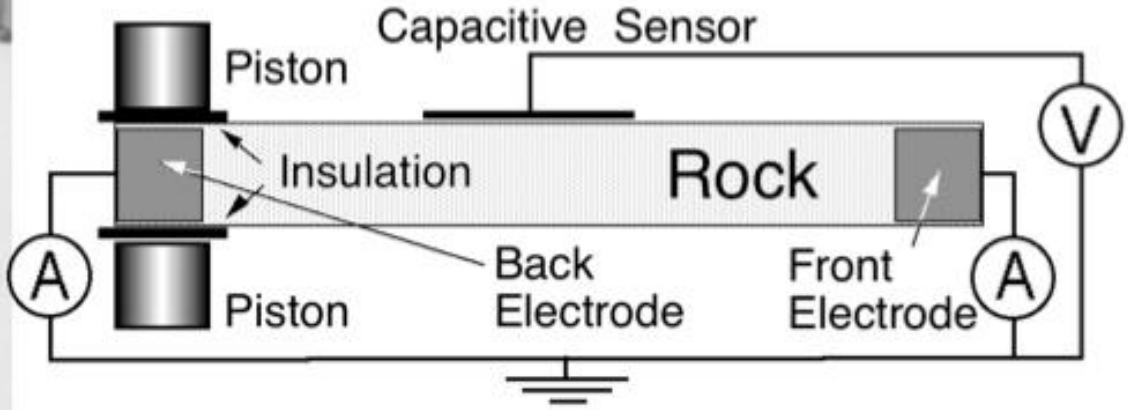
# Electrical Resistivity Values

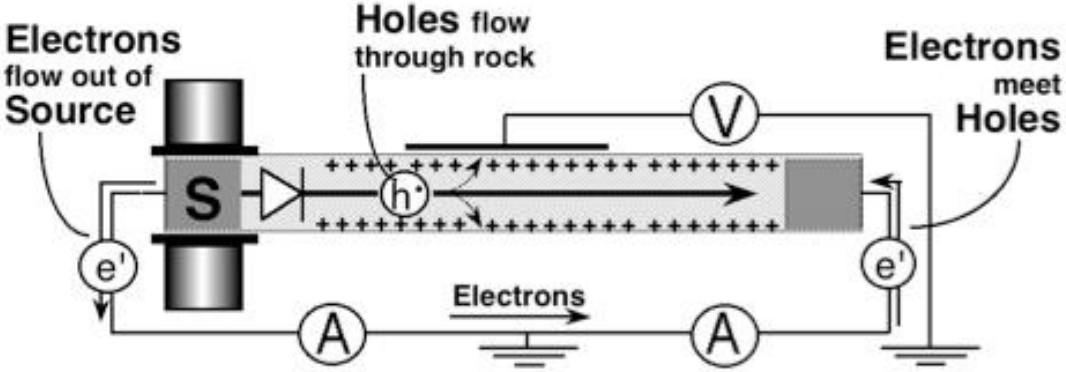
## Rocks, Soils and Waters*

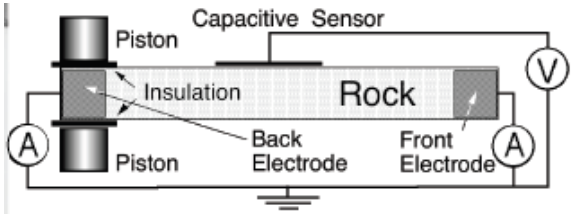


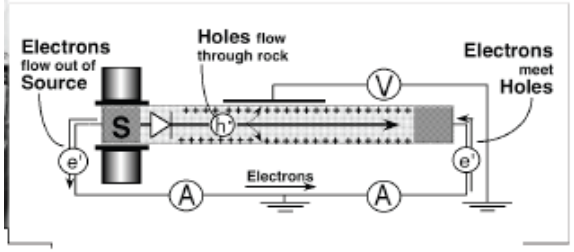

* Only approximate resistivity ranges can be given because the resisitivities of rocks are strongly influenced by fissuring, porosity, ground water conductivity and saturation.

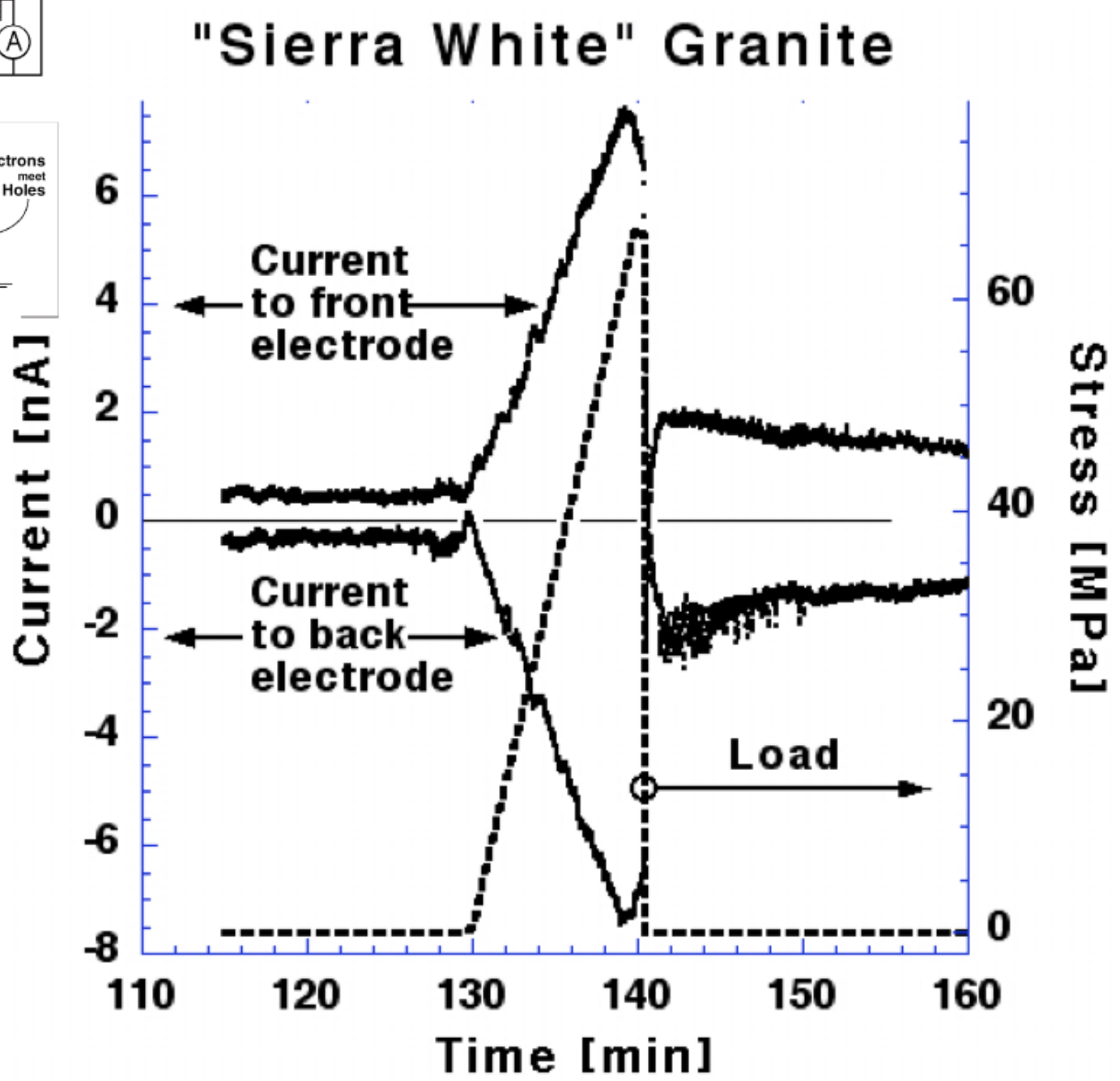


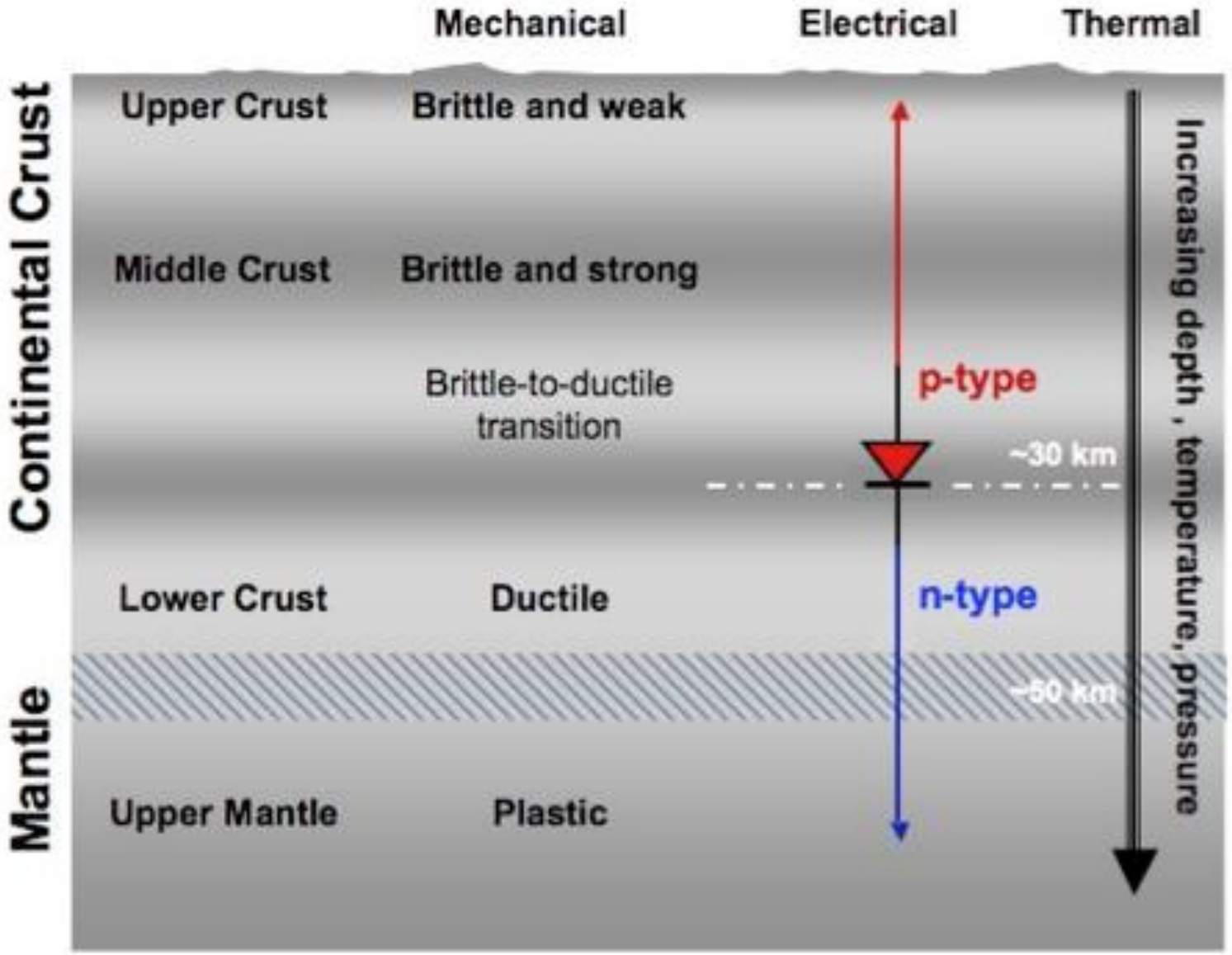

Capacitive Sensor
Piston
Insulation
Rock
V
A
Back
Electrode
Front
Electrode
A
Piston

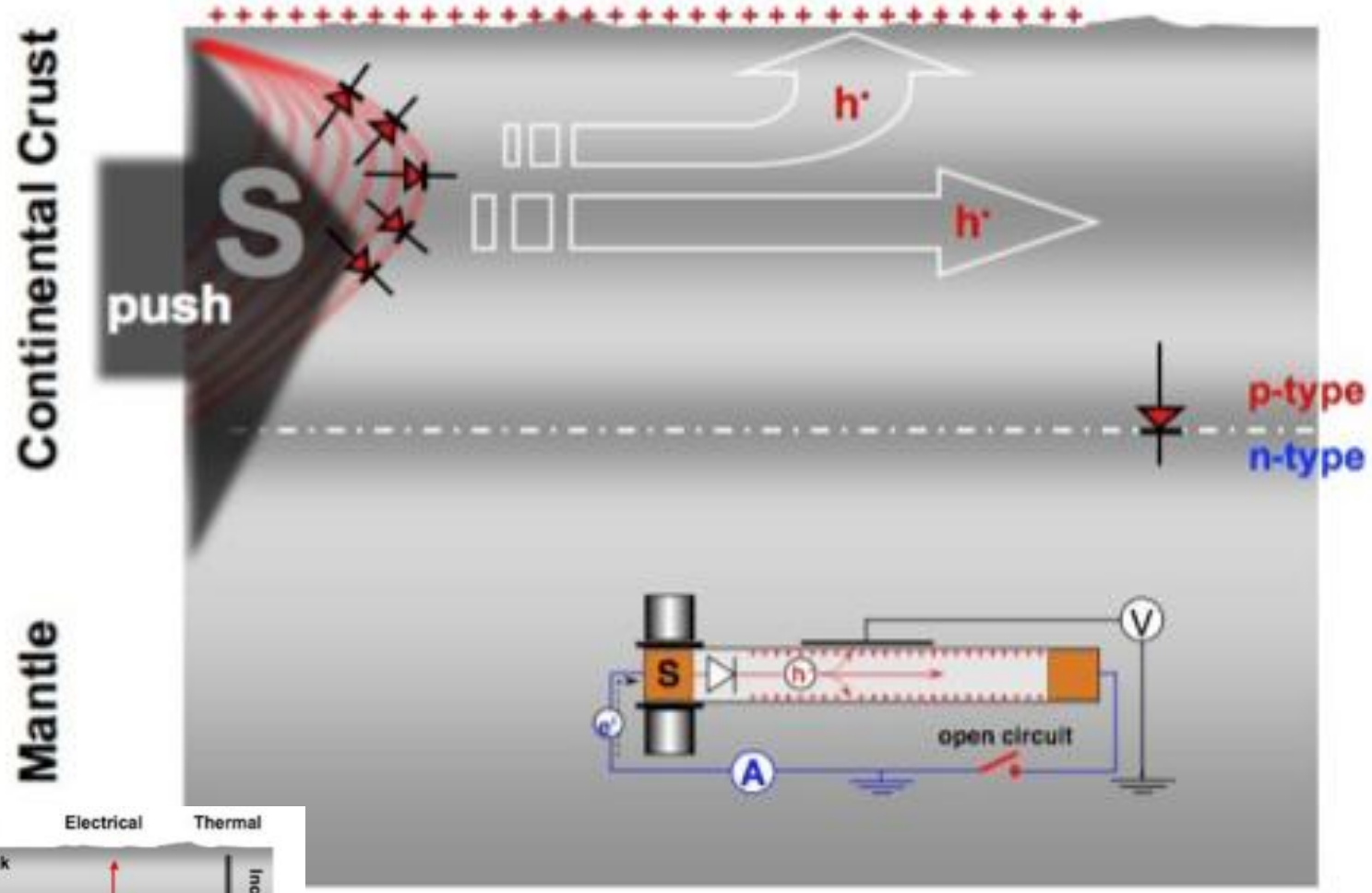

Electrons
flow out of
Source
Holes flow
through rock
Electrons
meet
Holes
V
S
h·
e'
e'
Electrons
A
A

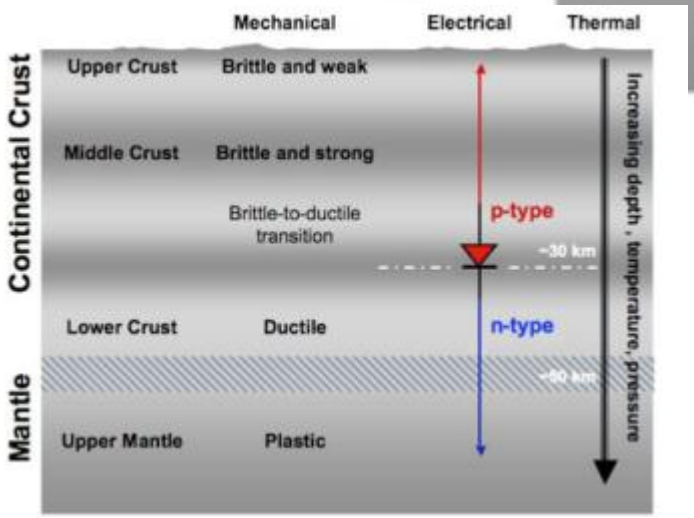

Capacitive Sensor
Piston
Insulation
Rock
V
A
Back
Electrode
Front
Electrode
A
Piston


Electrons
flow out of
Source
Holes flow
through rock
Electrons
meet
Holes
V
S
h+
e-
e-
Electrons
A
A

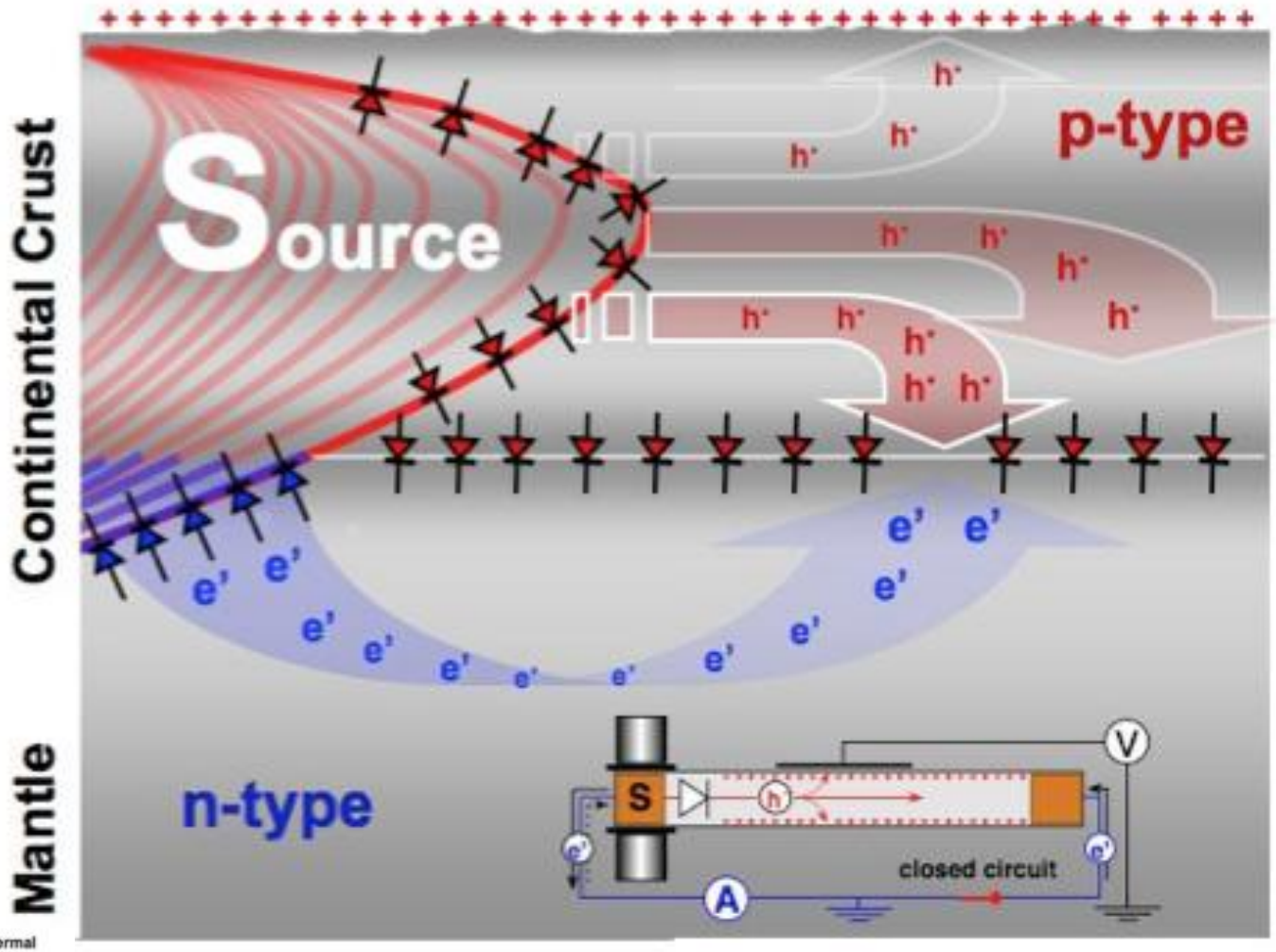

"Sierra White" Granite
Current [nA]
Stress [MPa]
Time [min]
Current
to front
electrode
Current
to back
electrode
Load
6
4
2
0
-2
-4
-6
-8
60
40
20
0
110
120
130
140
150
160

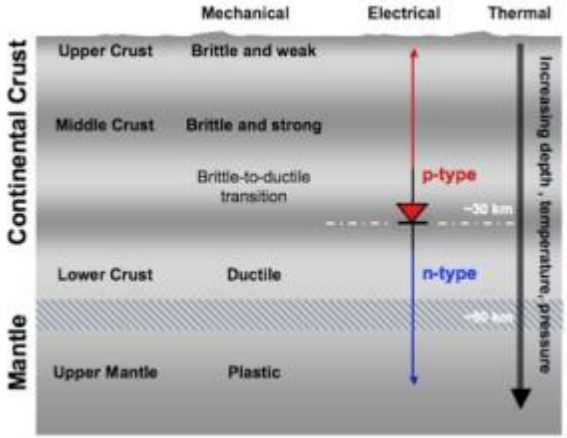

Mechanical
Electrical
Thermal
Continental Crust
Mantle
Upper Crust
Brittle and weak
Middle Crust
Brittle and strong
Brittle-to-ductile transition
p-type
~30 km
Lower Crust
Ductile
n-type
~50 km
Upper Mantle
Plastic
Increasing depth , temperature, pressure




**Case 1: Battery circuit open**




**Case 2: Battery circuit closed**

Maxwell :

$$\mathbf{B} = \mathbf{v} \times \frac{1}{c^2}\mathbf{E} \; ;$$

